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Review

Laser-induced Joining of Nanoscale Materials: Processing, Properties, and Applications

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Highlights

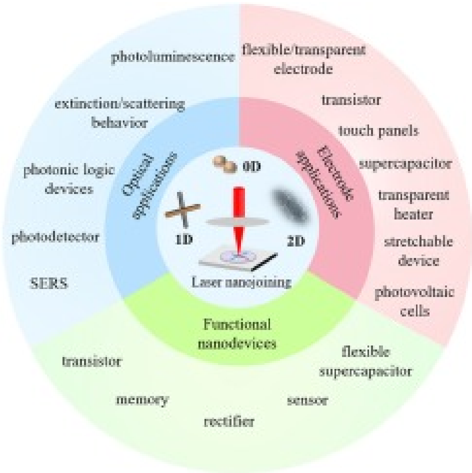
- Current progress of laser joining of multidimensional nanomaterials are reviewed.
- Laser joining of homogeneous and heterogenous nanomaterials are summarized.

- Mechanism of laser nanojoining and practical applications are discussed.
- Challenges and directions for future research in laser nanojoining are proposed.

Abstract

The rapid development of flexible and wearable nanodevices for a variety of commercial applications has identified a pressing need for targeted research on nanomaterials for use as building blocks in the fabrication of functional devices via bottom-up assembly. In the search for new fabrication technology, nanojoining or nanowelding has been selected as a promising method in this bottom-up approach. While there are a number of methods that can be used for nanojoining, laser processing is of interest because laser nanojoining combines high-precision property with a flexible manufacturing platform. In this review we discuss how this technology can be implemented in practical applications and outline the advantages and limitations of laser-induced nanojoining. We emphasize how laser nanojoining introduces reliability and reproducibility to the joining process, and show that this is due to precise control over heat-input in nanoscale dimensions. We also review the ways in which laser nanojoining can be integrated with other commercial fabrication operations in practical applications. To illustrate the flexibility of laser nanojoining, we give a detailed summary of joining involving a wide variety of multidimensional heterogeneous nanomaterials in the form of zero-dimension, one-dimension and two-dimension as well as the hybrid combination of these building blocks. We also discuss how laser processing can be used to generate hybrid materials with new functionalities as electrodes for various devices, optical systems and in functional circuits. The mechanisms and strategies in laser-induced joining of nanomaterials are systematically discussed in this paper along with a review of the properties and applications of the joined nanostructures. We also review some challenges in the implementation of this technology and discuss some possible directions for future research into laser-induced nanojoining.

Graphical abstract



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Keywords

Nanojoining; nanowelding; laser processing; homogeneous and heterogeneous junction; laser-matter interaction

Introduction

Nanoscale materials with different dimensionalities, including almost zero-dimensional nanoparticles, one-dimensional nanowire/nanotubes and ultrathin two-dimensional materials have many unique mechanical, electrical, thermal and optical properties compared to their corresponding bulk constituents. These prophorerties can be exploited by using low dimensionality components as building blocks in new technologies in the fields of electronics, photonics, energy generation and conservation, catalytic systems, and environmental monitoring, etc. For example, metallic nanoparticles and nanowire networks are providing low cost alternatives to the costly indium tin oxide (ITO) thin films traditionally used in transparent flexible electrodes [[1], [2], [3], [4]]. One-dimensional semiconductor nanowires such as Si, ZnO, and TiO_2 have been investigated for use in functional circuit components such as transistor [5,6], memory [[7], [8], [9]], photodetector [[10], [11], [12]], etc. However, the practical use of these nanoscale materials in devices has identified a number of significant challenges involving the integration of nano-scale components during system assembly. For example, despite their large intrinsic conductivity, the high contact resistance between individual metallic nanowires increases sheet resistance which is not an acceptable property in electrodes. Low adhesion strength between the nanowires and the underneath substrate is also disadvantageous because of possible mechanical failure during testing. In another example, the high surface-to-volume ratio of one-dimensional semiconductor nanowires and the ultrathin 2D materials makes devices containing these components extremely sensitive to surface condition [13] and introduces difficulties in engineering the nanowire or 2D material/electrode contact to ensure reproducible performance [[14], [15], [16]]. It is now apparent that control over the characteristics of the nano-contact is fundamental to applications of nano-engineered systems, especially in future functional nano-devices such as nanoelectromechanical system (NEMS) that require the joining of a variety of different nanocomponents and substructures of these components. Complex structures of this kind cannot be synthesized by direct chemical synthesis [17] or assembled by other methods [[18], [19], [20]].

In view of these issues, new technology is needed to enable the reliable assembly of nano-scale building blocks and nanocomponents for the assembly of functional nano-devices. As a result, the new field of nanojoining or nanowelding is becoming increasingly important as it facilitates the integration of individual components and enhances the mechanical strength of bonds between various structural elements while optimizing electrical and optical connectivity [21]. Nanojoining can lead to robust chemical bonding instead of the van der Waals or other weak bonding that often exists between nanomaterials [22,23]. A significant advantage of nanojoining is that this technique generally results in improvements in electrical conductivity and in the mechanical properties of the nanoscale materials involved in the assembly [24]. For example, a reduction in contact resistance among different nanoparticles or nanowires and improved adhesion to the substrate can be realized by different nanojoining methods as a reliable alternative for the next generation of flexible and transparent electrodes [25]. Furthermore, the joining of different nanoscale building blocks can directly and easily lead to the formation of complex shapes [26,27]. The joining process can also create structures with multiple individual building blocks each of which retain their intrinsic mechanical and optical properties [28]. It has also been found that the joining process can be controlled to engineer the atomic structure of any interfacial layers that exist between nanoscale materials, resulting in improvements in nanodevice performance based on one-dimensional nanomaterial [29,30] and 2D materials [[31], [32], [33]]. For example, theoretical studies show that the metal-semiconductor heterojunction between Pt and TiO_2 can transform from a Schottky to an Ohmic-type contact by varying the level of oxygen deficiency at the interface [34]. It is also found that the substitution of Au atoms at oxygen vacancy sites in TiO_2 increases the Schottky barrier at the Au/TiO_2 junction [35]. Experimental studies also suggest that the Schottky barrier between metal and semiconductor depends not only on the properties of the bulk materials, but also the interface structures such as crystal faces, surface terminations, the bonding pattern at the interface [[36], [37], [38], [39], [40], [41]]. These parameters can be controlled by tailoring the nanojoining conditions, providing a great opportunity for the engineering of the performance of the fabricated nanodevices. Current studies have demonstrated that the joining of dissimilar nanoscale materials to form heterojunctions can be further applied to nanoparticle-2D materials, nanoparticle-nanowire heterojunctions and nanowire-2D material heterojunctions [42,43]. The unique functionalities embedded in these structures by nanojoining can be applied to the development of a variety of active and/or passive mechanical, optical, electromagnetic and photocatalytic devices [44].

In general, the objective of all joining processes is to establish robust, stable connections between two or more individual components. These characteristics can be achieved in a variety of ways, but the dominant technologies for bonding of nanoscale structures have focussed on solid-state bonding, soldering/brazing and fusion bonding. Solid-state bonding includes joining techniques such as press bonding, diffusion bonding, and cold joining in which melting is absent. Solid state processes generally require an energy source to promote the diffusion of atoms across the joint boundary in the creation of a bond. In the solid-state bonding process, coalescence arises from the diffusion of atoms across the interface. This diffusion is enhanced as the scale size of the nanocomponents is decreased because of a lowering of the activation energy and an increase on specific surface energy. The overall effect is to reduce the bonding temperature and pressure. This has been seen in the use of Ag nanoparticles and nanowires to bond a substrate under low pressure [45] and low temperature conditions [46,47], facilitating the joining of heat sensitive, flexible substrates for electronic packaging. Another example of solid-state bonding is the application of ultrasonic welding to bond carbon nanotubes (CNTs) to Ti electrodes. The resulting Ti-C bonds created in the welding process can lead to the formation of Ohmic contacts with a corresponding improvement in the electrical performance of CNT based transistor devices [48]. A similar study has described the press bonding of ZnO [49] and Ag [50] nanowires on an electrode. This method can also be used for the joining of large scale Ag nanowire networks [51] and perovskite nanowire networks [52], greatly improving the performance of devices using this fabrication technique. Similarly, by fabricating pre-patterned nanowires on the substrate, homogenous and heterogeneous metal-metal and metal-metal oxide nanowire junctions can be formed in a large area with thermal pressing technique and various applications such as polarization color filters [53], photocatalyst [54], color electrode [55], gas sensing [56] and security identification [57] were realized. Some nanoscale building blocks can also be joined without pressing. It has been shown that ultrathin single crystal Au nanowires can be joined together by only mechanical contact at close to room temperature and that these joints have strengths comparable to that of the bulk material [58]. The realization of joining is attributed to the ultrasmall dimension of the nanoscale sample that permits oriented-attachment assisted through an enhancement in the surface-atom diffusion. A similar effect has been seen in the cold joining of tin oxide nanorods [59]. Recently, cold bonding of dissimilar nanoparticles at room temperature has also been observed. Grouchko et al. found that the wetting of Ag nanostructures can occur on contact with Au nanostructures at room temperature. Bonding under these conditions is induced by the activated migration of Ag atoms on the surface of Au nanoparticles and the formation of Ag-Au bonds [60]. Huang et al. found the joining phenomenon occurred between as synthesized Au nanoparticle and chalcogenide nanoparticle [61]. It was found that desorption of surface ligands induced conformal contact between the nanoparticles promoting the diffusion of Au into the chalcogenide material generating the bonds that act to join these dissimilar nanoparticles. These phenomena provide a deeper understanding into the role played by thermodynamics and interfacial nanostructures in cold-joining of nanoscale components.

Soldering/brazing is a method for joining materials whereby the base materials remain un-melted. This occurs when molten filler material is introduced between components to form a metallurgical bond. During the soldering/brazing process, dissolution, and diffusion of the solid-state material into the molten liquid in the joint is accompanied by the formation of a reactive region. Early research on joining of nanomaterials by soldering often required that a micromanipulator be used to precisely add soldering material to the junction prior to reflow or current induced joule heating and the formation of a robust joint. Examples include soldering on a single-layer graphene [62], the joining of two metallic [63], semiconductor nanowires [64] by using metallic solder. Recently, Cui et al. also applied optical fiber probe laser to heat up the Atomic Force Microscopy (AFM) probe tip functioning as the heating source for the nanospot welding of CNTs [65] and polystyrene nanoparticles [66]. However, obvious disadvantages of this technique are that only a single joint can be formed at one time and that the operation requires a complex accurate manipulation of the nanoscale components before joining can occur. Electrodeposition of multi-segmented metallic nanowire followed by a simple reflow process with an embedded nanoscale solder has been proposed for forming metallic nanowire networks as well as for joining CNTs [26,67,68]. The high complexity of this process, however, is a concern for high throughout. Recently, specific nanomaterials, including dry conducting polymer solution [69], sol-gel metal oxide nanoparticles [70] and chemical-synthesized Ag nanoparticles [[71], [72], [73]] have also been used as nano-solders for joining nanowire networks. These solutions are introduced into the joint by capillary forces and can nanosolder connections between Ag nanowires without any heat treatment, light irradiation or mechanical pressing. This method shows great promise in the fabrication of transparent electrodes combining low sheet resistance and high optical transmittance.

Unlike soldering or brazing that needs filler material, fusion welding involves melting at the contact surface between two materials creating a liquid that flows to fill the region within the joint. Fusion joining involves the input of enough localized heat for sufficient time to melt both components in the region of the joint. This heat input is typically provided by energy sources such as pulsed and continuous wave (CW) lasers, intense optical lamps, electron/ion beams, joule heating, plasma or microwave irradiation and chemical reactions. Deposition of this energy on the requisite timescale causes melting and is then followed by the establishment of a stable fusion zone. Thermal quenching then results in solidification and the formation of a bond between the two components. The processing window that defines the region of an acceptable bonding is strongly size dependent as nanoscale structures have significantly lower melting temperatures than those found in bulk materials. In addition, the latent heat of fusion in nanoscale materials is often smaller than that measured in the bulk so that melting is facilitated. Fusion welding in nanojoining has many advantages compared to soldering and brazing because no filler is required, and the components can be assembled without micromanipulation. Fusion joining has been studied in a wide range of nanoscale bonding configurations involving many different materials. A summary of non-laser based fusion joining methods for some metallic, semiconductor, and polymer nanoscale components, together with the formation of heterojunctions between different nanoscale materials is given in Table 1.

Table 1. Summary of nanoscale materials joining by different fusion joining methods.

	Electron/ion beam	**Optical method (excluded laser)**	**Joule heating**	**Chemical reaction/capillary force**	**Heat treatment**	**Other energy sources (Plasma, micro-wave, etc.)**
Metallic	[74]	[25,[75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86]]	[24,[87], [88], [89]]	[[90], [91], [92], [93], [94], [95], [96]]	[97,98]	[99,100]
Semiconductor	[[101], [102], [103]]			[104]	[105]	

	Electron/ion beam	Optical method (excluded laser)	Joule heating	Chemical reaction/capillary force	Heat treatment	Other energy sources (Plasma, micro-wave, etc.)
Ceramic	[[106], [107], [108], [109]]			[108]		
Polymer		[110]				
Heterojunction of materials	[74,[111], [112], [113]]	[[114], [115], [116], [117]]			[118]	[119]

The use of laser radiation for nanojoining, particularly irradiation with short and ultra-short laser pulses, has attracted much attention over the last two decades. In the context of other available technology, laser irradiation has many advantages. These include reliability, reproducibility, high-precision processing, and compatibility with flexible manufacturing systems. The localized heat input in laser welding makes it appealing for the joining of macro and micro scale components in automotive, aerospace, medical and microelectronics applications. Compared to other techniques for nanojoining, laser-induced nanojoining has the advantage of non-contact processing, high throughput and operational efficiency. The laser nanojoining process can also be integrated with other operations such as roll-to-roll, screen printing and ink-jet printing processes for the mass production of e.g. flexible transparent electrode fabricated with metallic nanowire networks and sintering of metallic nanoparticles. These advantages make laser-induced nanojoining very attractive for joining of nanomaterials either as free-standing structures or as components on heat sensitive substrates. Laser joining is also well suited for the assembly of systems affected by excessive heat input, require precision joining or susceptible to distortion as found, for example, in the manufacture of flexible and wearable devices. The localized heat affected region produced during laser nanojoining is also compatible with applications that require selective patterning on a conductive substrate or localized interface engineering of nanoscale heterojunctions between dissimilar materials.

This review focuses on describing the status of research on laser-induced nanojoining for a variety of different types of nanomaterial. We also discuss the nature of mechanisms that are important in laser-induced joining, properties of the joints produced and corresponding applications. After a general introduction to laser nanojoining emphasizing experimental methods and related joining mechanisms, this review outlines the current results on laser-induced joining of nanomaterials having different dimensionality. These include zero-dimensional nanoparticles, one-dimensional nanowires, and two-dimensional ultrathin materials in homogeneous and heterogeneous forms. We also review some practical applications of nanojoining including the fabrication of electrodes, the creation of functional circuit and optical elements. Finally, we provide some insight into possible future research directions in the new field of laser nanojoining.

Introduction to laser-induced nanojoining

Overview of laser-induced nanojoining

Since the first demonstration of laser emission by Maiman in 1960 [120], the laser has become an indispensable tool in many areas of science and technology. The usefulness of the laser derives from its ability to generate an intense collimated beam of monochromatic, coherent, polarized light. These characteristics make the laser an ideal tool for non-contact, localized material processing. Various types of laser with different output parameters have been developed based on stimulated emission from CO_2 gas, neodymium-doped yttrium aluminum garnet (Nd:YAG), various fibers, semiconductor diodes, mixtures of halogen and inert gases, and Ti: sapphire solids. These provide us with a wide range of options with respect to output power, wavelength, pulse duration and repetition frequency and have ensured that lasers are now extensively used in many important industrial applications, including aerospace, the construction industry, industrial robotic systems, and automotive manufacture to name a few. The non-contact nature of laser processing is well suited for applications that require high power density, spatial focussing and directed heat input. The high intensity in a focussed laser beam means that all known materials can be easily melted ensuring that laser radiation is ideal as a remote joining/welding source. In the industry, laser processing is now widely used for macro-welding of similar and dissimilar materials. Examples include, but are not limited to, the macrojoining of steel, aluminum, titanium, Kovar and nickel, microjoining of Ni-Ti microwires for medical applications, and the formation of interconnects in microelectronic interconnection [[121], [122], [123], [124]].

The first studies on laser joining of nanomaterials began in 2003 when the dissimilar joining of Pt and Au nanoparticles in solution was realized using a nanosecond Nd:YAG laser [125]. This has now been followed by many other studies as laser-induced nanojoining has been demonstrated to be a viable technology. Research activities can be roughly divided into two distinct stages as indicated by the timeline tracing the evolution of laser-induced nanojoining shown in Fig. 1. Before 2012, activity on laser and light induced nanojoining is seen to be of limited scope, as only a few reports were published on laser-induced nanojoining. Little information was available on the physical mechanisms involved in nanojoining by laser irradiation. This situation changed in 2012 when the first study of joining Ag nanowire using tungsten-halogen lamp was reported [77]. Since that time, there has been a surge of investigations on laser-induced nanojoining covering many more different types of materials and with different types of laser sources. An important aspect of this activity includes studies of the joining of metallic nanoparticles as well as metallic nanowires. This emphasis has been largely driven by the possibility of using sintered metallic nanoparticles as a bonding layer and the use of metallic nanowires in transparent electrodes [77]. Joining of nanoscale semiconductor materials has also emerged for applications in dye-sensitized solar cells. In addition, a better understanding of the laser-nanomaterial matter interaction is now available and there are comprehensive theoretical and experimental studies of thermal effects induced by laser radiation in a variety of different nanomaterials [[126], [127], [128]]. Molecular dynamic simulations that follow the evolution of the joining mechanism on an atomic basis are also available and have supported these experimental studies [129]. Features such as high-power density and the wide range of operational parameters with lasers have greatly expanded the range of nanomaterials that can now be joined. Furthermore, these new capabilities have enabled laser joining of new and functional nanomaterials including metal oxide [130,131], semiconductor nanomaterials [132] and 2D materials such as graphene [133]. Joining of different nanomaterials to create heterojunctions is also now feasible. The latest work has shown that laser processing can be used to construct metal-semiconductor heterojunctions [134,135], and engineer contacts of semiconductor nanowires on an electrode for inclusion in operational circuit elements [[136], [137], [138]]. Other research on nanojoining using conventional light sources such as halogen lamps, flashlamps and xenon lamps is still very limited. The generation of heat within nanomaterials with these sources is limited by low optical intensities and difficulties in the tight focussing of incoherent light beams. Most research for nanojoining by these types of lights are focusing on metallic nanomaterials.

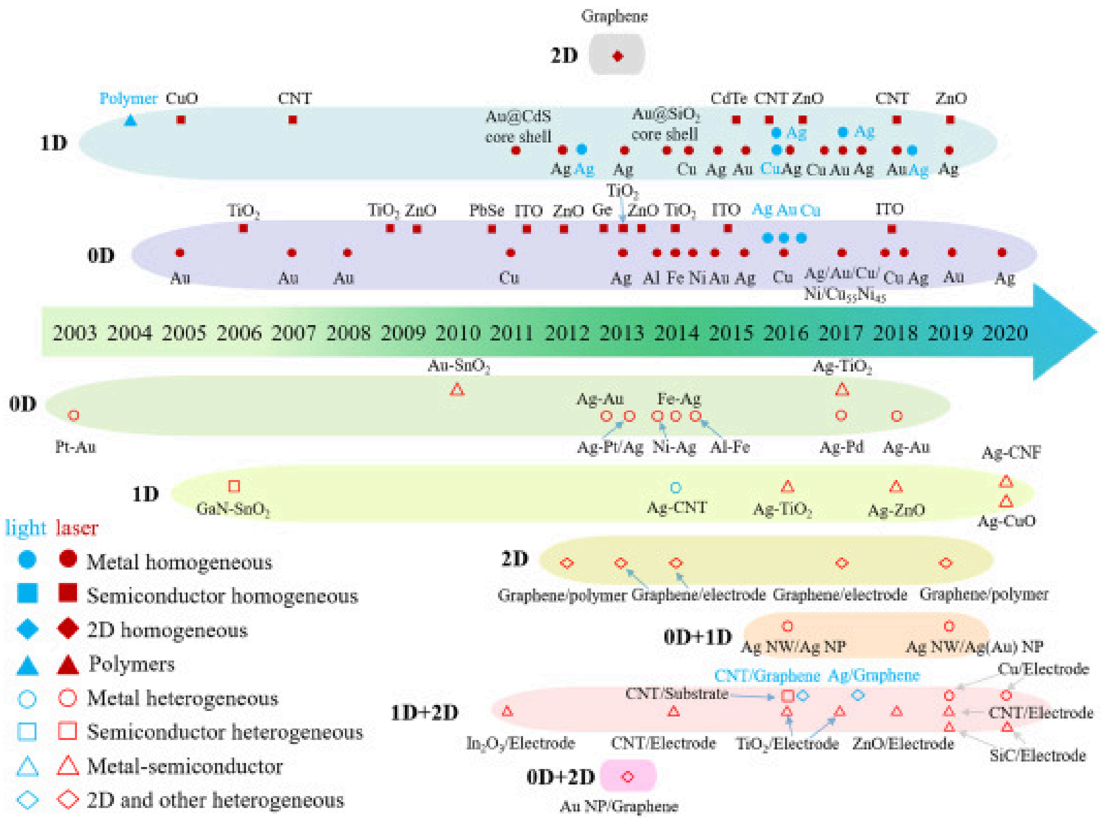

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Fig. 1. Timeline for research on laser-induced nanojoining. 0D, zero-dimensional nanoparticle, 1D, one dimensional nanowire/nanotube/nanorod. 2D, two-dimensional material. For comparison, nanojoining achieved by irradiation with incoherent light are also included with the caption are colored blue.

It should be noted another type of laser processing, pulsed laser deposition (PLD) for the formation of nanomaterials on substrate have also been widely studied. During the nanoparticle or nanocolumn formation on substrates by PLD process, the targets are ablated first and evaporants are generated that will deposit on the substrates. Upon on the substrates, the atoms of the targets (metals for example) will nucleate first randomly on the substrate. Further attaching atoms to the nucleation sites lead to the growth of the nucleation sites until a critical size then is followed by coalescence with coarsening for larger coverage. Most of the reported research on metallic nanocrystals formation in the nanocomposite on dielectric substrates by PLD is based on the Volmer-Weber growth mechanism, in which three-dimensional islands are formed with adatoms attaching to each other and further nucleation and growth lead to the increase of these nanocrystals [[139], [140], [141]]. It was also reported that the implantation of metallic layer into the dielectric substrate layer could occur due to the high kinetic energy of metallic atoms from the laser ablation process [142]. Furthermore, the following deposition of dielectric layer will first cover the space between these three-dimensional islands, then completely bury them, forming the structure in which metallic nanocrystals are embedding in dielectric host. Even though it is a laser related process, we could not classify it as laser-induced nanojoining because no specific nanomaterials are involved in this process and the deposits have undergone a complete process of ablation, nucleation, growth and further coalescences. This is more likely to be laser-induced nanostructure formation instead of laser-induced nanojoining. In our review, laser-induced nanojoining is a process that laser irradiation of nanomaterials that are adjacent or in proximity therefore the photon energy from laser irradiation can cause the joining of these materials.

Laser-induced nanojoining experimental setup

Most of reported experimental configurations for laser-induced nanojoining fall into the four general categories shown in Fig. 2. These consist of techniques for large area irradiation, laser focusing, beam scanning and laser processing in a liquid.

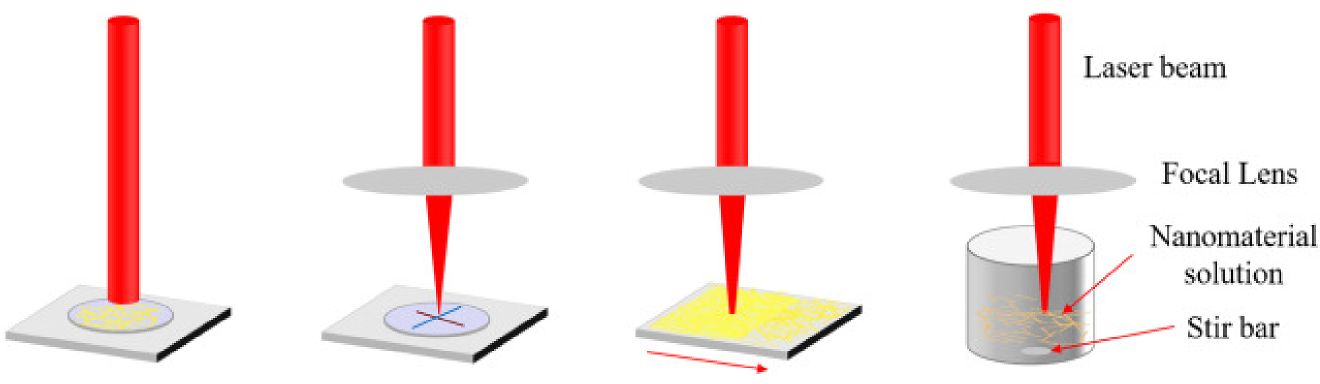

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Fig. 2. Various configurations used for laser-induced nanojoining (a) large area irradiation, (b) laser focusing, (c) laser scanning or rastering and (d) laser processing in a liquid.

Large area irradiation

The most direct approach for nanojoining involves the expansion of the laser beam to cover the complete area of an array of components deposited on a surface [143]. This simple irradiation technique is easily implemented and does not require any pre-positioning of individual components. Large area irradiation has been widely adopted for the joining of nanoparticles [[144], [145], [146]] and nanowires [136,147,148] under a variety of conditions. The intensity of the irradiation and fluence at the surface of the sample are readily controlled by insertion of neutral density filters in the incident laser beam or by the combined use of a half-wave plate and a linear polarizer, while the irradiation time can be adjusted by passing the beam through a mechanical or electrical shutter. The effect of laser parameters such as polarization, laser power and laser irradiation time on joining behavior can be easily mapped out in this experimental configuration.

Laser focusing

Compared to large area laser irradiation, laser focusing generally increases the power density (W/m^2) and fluence (J/m^2) in the irradiation zone. The spot size is determined by the input beam diameter and the method for focusing the beam. Normally, a combination of a focal lens and an objective lens with different magnifications are used for focusing the beam. The smaller the beam size, the higher the laser power density in the working region. Unlike with large area irradiation, the focussed beam can be directed to specific locations such as the junction between nanowires or the contact region between a nanowire and an electrode, minimizing collateral exposure in the surrounding area. This focusing method is used for spot nano-joining of nanowires [[149], [150], [151]] as well for contact engineering in specific regions between a semiconductor nanowire/2D graphene and an electrode [30,133]. However, the downside of this method compared to large area processing is that it sacrifices throughput for precision.

Laser scanning

Laser scanning combines the advantages of large area processing with tight focusing. The samples to be joined are mounted on a two-axis stage in which the scan speed is controlled by the movement of the stage. Alternatively, the components can be kept stationary while the focussed laser beam is rastered by a mirror-based scanning system over the samples. Laser scanning is often used for the joining as well as sintering of large area of metallic nanowires and nanoparticles. It can be directly used for the fabrication of large area electrodes and can be integrated with other processes such as roll-to-roll printing [129] and ink-jet printing [152]. The scanning speed can be >3000 mm/s, as required for the fabrication of large area electrodes [153]. An advantage of high scan speed is that it can inhibit unwanted thermal effects such as heating damage to the substrate [129] or thermal oxidation of metallic nanoscale materials such as Cu nanowires [154]. Furthermore, laser scanning can be used for patterning and the production of unique structures. This is easily accomplished by varying laser scanning parameters, such as pixel size, overlap of pixels and dwell time at each location when using stepping. For example, laser scanning can selectively write metallic electrodes by sintering a metallic nanoparticle solution deposited on a substrate. This is then followed by a second washing process that reveals the desired electrode pattern. This laser based technique eliminates costly, time-consuming photolithography and gives a direct-writing patterning capability [155]. When using a high magnification objective lens with the XY stage, femtosecond laser-induced patterning having a line resolution down to 380 nm can be obtained [156].

Laser processing in a liquid

Laser processing in liquid is a clean and rapid process for the formation of nanostructures with various compositions and morphologies [157]. Recently, researchers applied this method for the joining of homogenous/heterogenous nanomaterials in solution. In this process, a solution of nanoparticles with known concentration is first dispersed in a solvent such as water or ethanol. For joining of dissimilar materials, two or more types of nanomaterial are dispersed in solution with a controlled concentration ratio. The mixture is then irradiated under tight focussing conditions for a fixed period of time. This ensures that the irradiation volume contains a small number of nanoscale materials. Focusing the laser beam also increases the power density and enhances energy deposition, measured by calculation of the fluence. Some researchers also used non-focused laser beam for irradiation [43,158]. It has been found that changing the solvent can influence the morphology of the nanomaterials subjected to laser processing, so this factor must be addressed in experimental design. A stir bar is normally introduced to make sure that the nanoscale materials are homogenously distributed so that the laser irradiation effect is uniform. Laser irradiation in a liquid has shown great potential for the homogeneous [143,159] and heterogeneous [125,[160], [161], [162]] metallic nanoparticle joining and metal-semiconductor nanoparticles joining [158] and is expected to be feasible for the joining of other types of nanomaterials. However, a main disadvantage of this method is the low productivity for heterogeneous nanomaterials joining. Furthermore, many nanowires and nanoparticles are passivated by coating with an organic layer to prevent aggregation in solution, and this material is also subject to modification during laser processing in liquid and will have an impact on the joining behavior.

Mechanism for laser-induced nanojoining

Laser-matter interaction

In all laser processing applications, including the joining of nanomaterials, the initial stage consists of the absorption of individual photons by atoms or molecules at specific locations within the material. The nature of this absorption will be determined by the composition of the material and the photon wavelength. Following absorption, some of the photon energy may result in photochemical dissociation or the excitation of electrons to higher energy levels within the material. The breaking of chemical bonds in oxides and semiconductors can lead to the formation of defect centers and the evolution of volatile constituents such as oxygen while the excitation of electrons in metals may result in photoemission. In many cases, most of the excitation energy is retained within the nanomaterial in the form of electron-hole pairs, plasmons and super-heated electrons that eventually contribute to thermal heating of the material. We outline the role played by these phenomena in the rest of this section.

At low incident laser intensity, single photon absorption is the dominant mechanism for the conversion of photon energy into thermal heating. For example, single photon absorption excites valence electrons into the conduction band when the photon energy exceeds the bandgap energy in a direct bandgap semiconductor. In the case of a semiconductor such as silicon with an indirect bandgap, the process involves the absorption of a single photon with energy greater than the bandgap energy together with creation of a phonon that allows the excited electron to relax to the bottom of the conduction band. Multiphoton absorption can also play a role if the direct gap is greater than the photon energy or if the absorption of single photons absorption is inhibited by band filling. In metallic nanostructures, single photon absorption results in the excitation of electrons within the conduction band. Thermal equilibrium with the lattice occurs when the excited electron gas transfers its excess energy to the lattice. This process occurs on a femtosecond to picosecond timescale. Collective electronic resonances that involve the motion of the electron gas relative to the ion lattice are known as plasmons. Plasmons localized in surface states (surface plasmons) can also be excited by photo-absorption but the wavelength at which the resonance occurs is strongly dependent on the shape of the metallic structure, surface coatings and the proximity of other metallic structures. Resonance between the plasma absorption frequency and the photon frequency leads to an enormous enhancement in the coupling of light into the nanostructure. Localization and enhancement of electromagnetic field occurred in these nanomaterials due to the excitation of surface plasmons contributes to the localized heating and further joining of metallic nanostructures [163]. The properties of surface plasmon resonances and their use in metallic-related nanojoining will be analyzed in the next section.

The transition from the photoexcitation of carriers to the appearance of heat in the irradiated structure proceeds through a range of intermediate steps as the system relaxes to a thermal equilibrium state [164,165]. These different interactions and the corresponding time scales are summarized in Fig. 3. For irradiation of a metallic nanocomponent, the approach to thermal equilibrium can be divided into three distinct stages which depend on the laser pulse duration. In the first stage, photons are absorbed by the electron gas creating an excited state that thermalizes on a timescale of ~ 100 fs to a Fermi-Dirac distribution. This process occurs without energy transfer to the lattice so that the system can be initially characterized by two temperatures: one for the electron gas and another for the lattice. During this period, the combined system (electron gas and lattice) is in a state of non-equilibrium as the effective temperature of the electronic gas has increased while the temperature of the lattice (phonons) remains unchanged. Non-thermal structural changes and lattice softening can be initiated in response to the non-equilibrium between the lattice and the electron gas [[166], [167], [168]]. The second stage corresponds to electron-phonon thermalization which is accompanied by transfer of the excess heat contained in the electron gas to the lattice. The lattice temperature then increases as a result of electron-phonon scattering, producing a thermal equilibrium between the electron gas and the lattice over a time scale of tens of picoseconds. At this point, the lattice comes into local thermodynamic equilibrium in the structure, but the internal temperature may vary at various locations because it reflects the original distribution of the optical electric field and plasmon generation within the structure during excitation. This results in the appearance of “hot-spots” that are often responsible for melting and other non-equilibrium effects at surface sites and junctions. The third stage in the approach to equilibrium occurs as the structure transfers heat to the surrounding medium. This process can be characterized by macroscopic thermodynamic parameters such as the thermal diffusivity. Overall cooling rates also depend on particle size and morphology and the nature of the surrounding medium. For example, one calculation indicates that the steady state condition after excitation for spherical nanoparticles with diameters 10 nm, 100 nm and 1 μm is reached after ~ 0.1 ns, ~ 10 ns and ~ 1μs, respectively [126].

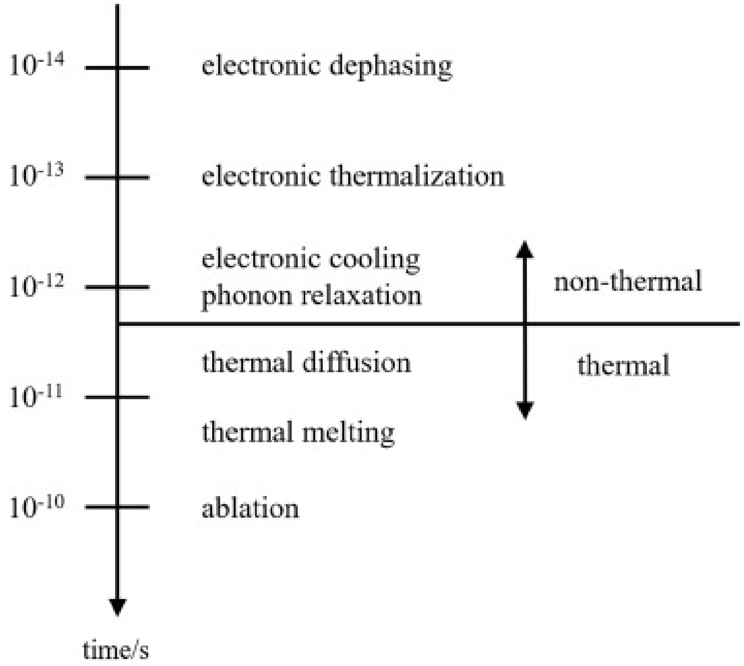

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Fig. 3. Timescales for various secondary processes in laser-matter interaction. Reproduced with permission from Ref. [165]. Copyright 1997, Elsevier.

From the above discussion, it can be seen that to consider the effect of various types of laser for the application of nanojoining, the duration of the laser pulses and the laser wavelength are key factors. Both CW and pulsed lasers are currently used for research and applications in laser-induced nanojoining. Pulsed lasers can be further divided according to pulse duration and repetition frequency with the shortest pulses extending into the picosecond and sub-picosecond (ultrafast) range. Irradiation with ultrafast laser pulses offers the possibility of producing non-thermal structural transitions since the laser pulse duration is shorter than the thermal coupling time between the electron gas and lattice phonons. In this regime it is found that the evolution of the temperature depends on both electron and phonon dynamics as described by the two-temperature model [126,167]. In contrast, excitation with longer laser pulses or CW laser radiation leads to thermal heating as described by classical heat transfer models [127,164,169]. In this model, the laser beam acts as a conventional heat source. The intensity of this source, and the overall heating efficiency, depends on pulse duration and temporal profile. While a single laser pulse can result in a rapid increase in temperature, the timescale for cooling following excitation is a strong function of the thermal diffusivity and geometry of the material. It is generally found that a rapid pulse repetition rate can result in a rise in baseline temperature when the interval between pulses is smaller than the cooling timescale.

A summary of laser joining of nanomaterials under a variety of laser irradiation conditions is given in Fig. 4. It should be mentioned that the cost of lasers and the availability of specific wavelength or pulse duration are also factors that may impact the corresponding research. This summary shows that CW lasers with different wavelengths are widely used for joining a range of homogeneous and heterogeneous nanomaterials. This might also be due to the wide availability of different types of CW lasers in the laser market. As a source for constant heating, it offers significant thermal energy distribution along the whole nanostructure and is suitable for large area sintering of metallic and semiconductor nanoparticles with a large processing window and more uniform product quality. Parameters outside of the processing window can cause the delamination of nanostructures when the energy input is too low, or the creation of damage in the underlying substrate when the energy input is too high [170]. In addition, the large heat affected zone associated with CW laser irradiation limits the production of high-resolution laser-patterned nanostructures or localized nano-joining on heat-sensitive substrates. High quality nanojoining using CW irradiation also requires higher average power than needed with pulsed lasers [171].

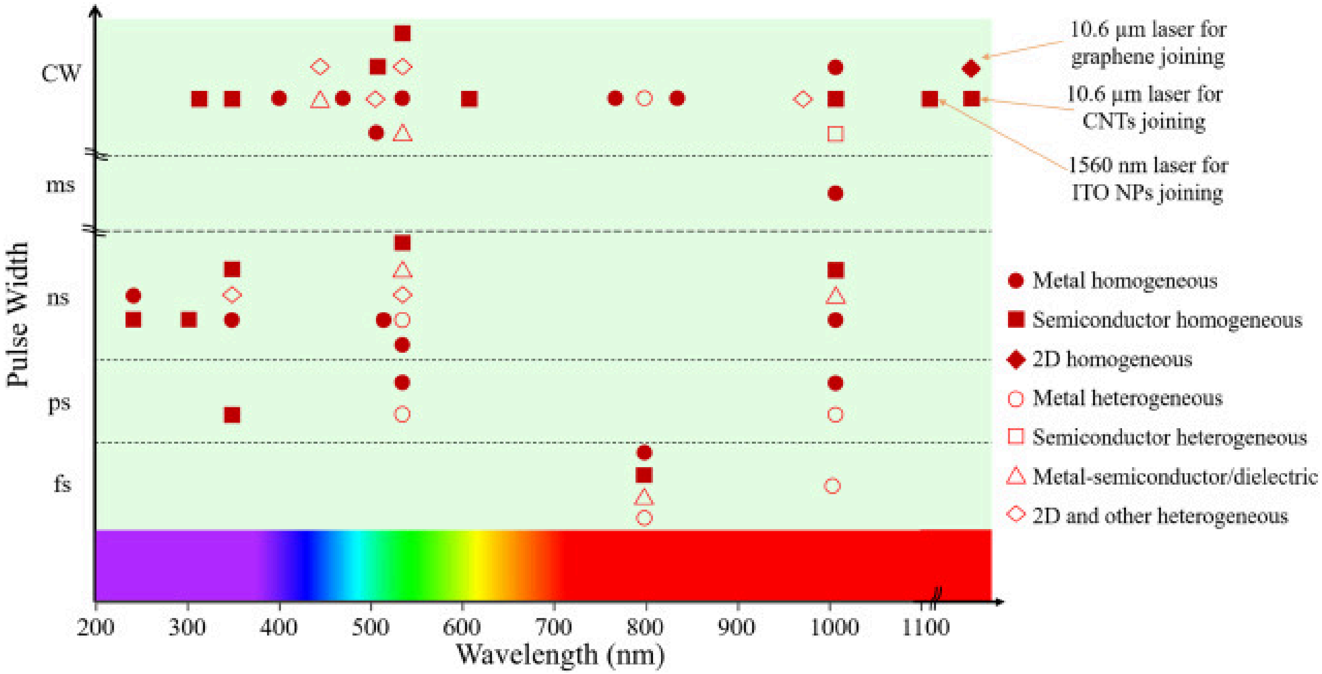

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Fig. 4. Summary of different types of lasers in perspective of pulse duration and wavelength for laser-induced nanojoining.

Pulsed lasers with a pulse duration exceeding several picoseconds have also been used for nanojoining of a variety of nanomaterials. As noted, pulsed laser irradiation provides higher laser heating efficiency and a smaller heat affected zone as well as processing at lower average power compared to CW irradiation. In practice, pulsed laser sources are effective in the creation of small features on a substrate when the laser scanning speed, laser fluence and laser repetition rate are optimized. Under these conditions, thermal damage to the substrate as well as to the surrounding area can be significantly decreased compared to CW laser.

Ultrafast lasers are characterized by very high energy intensity, and small pulse duration leading to minimal heat affected zone and low damage. The minimum heat affected zone makes ultrafast lasers promising for highly localized joining and localized laser processing. For example, different researchers have used fs laser pulses to locally process the interface of one-dimensional nanomaterials on electrodes, providing a promising method for interface engineering and further control over the performance of the corresponding devices [30,136,137,172,173]. Under the high intensity irradiation possible with fs lasers, non-linear processes such as multiphoton absorption can dominate interactions with ceramic, semiconductor and insulating materials. This enables laser processing of nano-materials with wide bandgap, high melting point and high damage threshold where no surface plasmon resonance is available [151,174].

Surface plasmon resonance for metallic nanoscale joining

When metal nanostructures such as nanoparticles and nanowires are irradiated at laser wavelengths within the absorption band corresponding to a localized surface plasmon resonance (LSPR), the giant optical resonance efficiently couples incident photon energy into the electron gas. The extent to which this occurs depends on the nature of the metallic nanomaterial as well as the morphology and size of the nanostructures [126,175]. The LSPR produces an antenna effect resulting in light collection from an area that is larger than the physical size of the component. This generates an intense electric field localized at the particle surface which produces hotspots at locations in the nanostructure where the plasmon energy is trapped [163]. Under these conditions the electron temperature greatly exceeds that of the lattice until energy is transferred from the laser generated plasmon into the manifold of phonon states. As the trapping of plasmons is enhanced at geometrical discontinuities such as constrictions and interfaces within the nanostructure much of the energy contained in the plasmon is deposited at these locations, leading to localized heating, softening and melting. This effect makes it possible to selectively melt material in the region adjacent to the interface between two or more nanocomponents. A calculated study shows that illumination with a fluence of 100 $\mu J/cm^2$ per pulse would lead to a temperature increment of 550 K for a single Au nanorod [176,177]. The characteristic dimensions of this selectively heated region can be much smaller than the laser wavelength, so that the production of plasmons in a nanostructure enables bonding over scale sizes much smaller than that predicted from the incident wavelength. This is a major advantage in the assembly of nanocomponents without collateral heating of surrounding structures.

In laser-induced joining of nanomaterials, individual nanoscale building blocks must be in contact or in close proximity prior to laser irradiation. Under these conditions, plasmonic coupling is strongly dependent on interparticle interactions as the near-field electric field distribution from one particle interacts with the fields on neighbouring particles and structures. Irradiation then produces a system exhibiting a coupled plasmonic response so that each component is influenced by the excitation state of surrounding nanostructures. Coupling of the electric field from the plasmons in one nanoparticle or nanowire to the electrons in an adjacent nanocomponent is a strong function of the gap or distance between the components [178,179], as well as the direction of laser polarization [180] and the orientation of the contact between the structures [148]. The plasmonic interactions trigger photothermal effects where electronic oscillations at the material surface are converted to heat, which, assisted by high laser fluence, raises the temperature that can lead to the joining of nanoscale materials [177]. A dominant parameter in all cases is the laser fluence which is the product of the laser intensity and the time duration of irradiation. The fluence is then useful as a measure of total exposure and can be used to characterize the range of acceptable irradiation conditions. For example, irradiation under low fluence conditions consisting of high laser intensity for a short time interval can result in the production of phenomena associated with the freezing in of non-equilibrium states in the absence of substantial thermal heating of the overall structure. On the other hand, irradiation under low fluence conditions consisting of low laser intensity for a long period of time will show evidence for thermal heating of the entire structure.

Numerical simulation has been widely used for the study of the LSPR effect that involves metallic nanomaterials. Electric field enhancement at the junction or contact, which is related to the laser polarization, laser wavelength and gap or overlapping distance is generally much higher than other regions due to the LSPR effect, as shown in Fig. 5. High temperature raise is expected in these regions that could potentially lead to the nanojoining. Temperature distribution in laser illuminated complex metallic nanostructures can be calculated numerically by combining the electromagnetic field distribution and heat diffusion equations under stationary conditions for both fs laser and longer pulse duration as well as in a dynamic way [127,181]. Steady-state local thermal imaging of metallic nanowire structures under laser illumination was experimentally obtained by high-resolution interferometry, providing direct evidence for the LSPR effect [128].

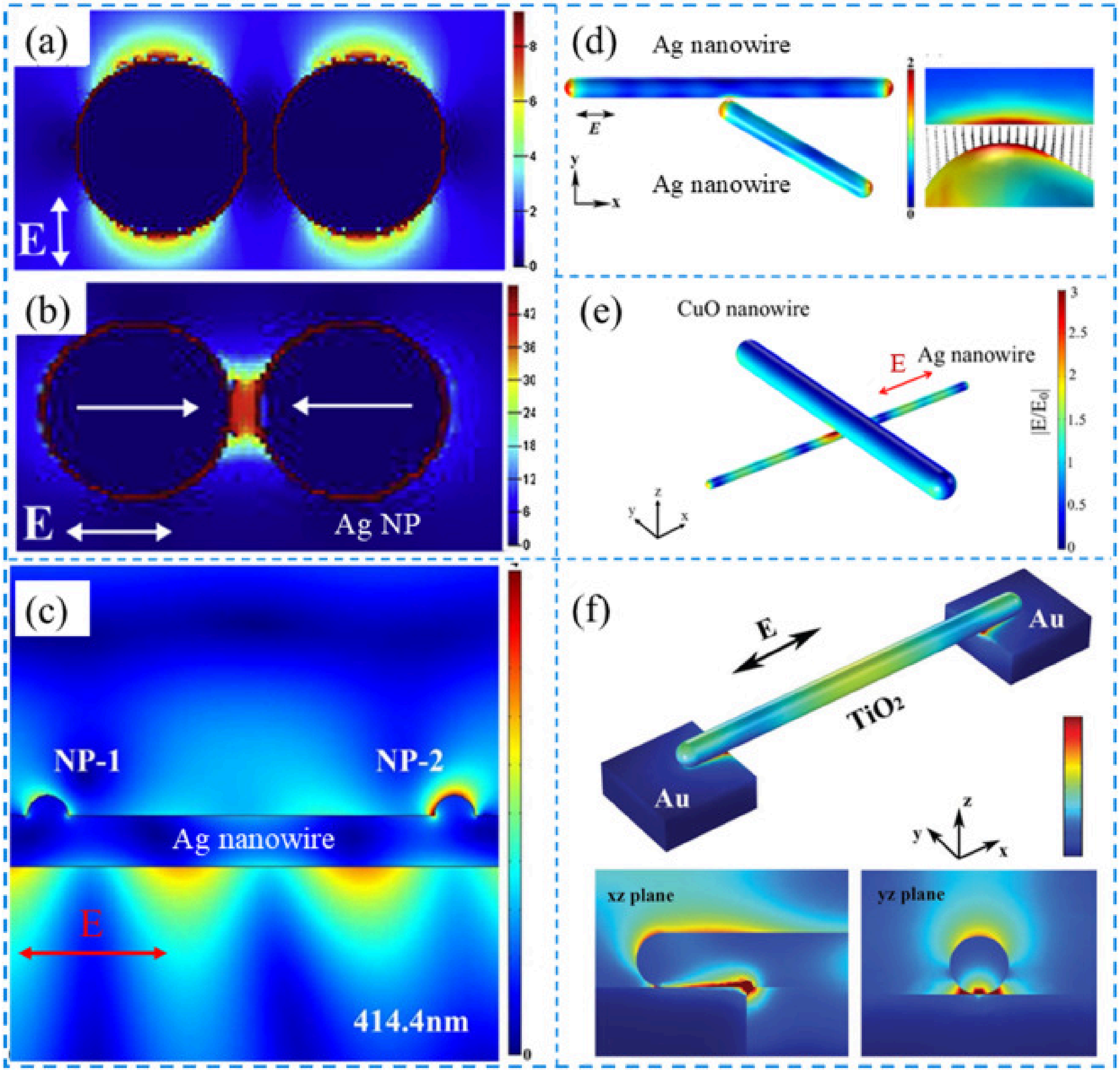

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Fig. 5. Numerical simulation for LSPR effect for the laser-induced nanojoining of different nanoscale building blocks. (a, b) metallic (Ag) nanoparticles joining with different polarizations. (c) Metallic (Ag) nanowire joining with metallic (Ag) nanoparticles. (d) metallic (Ag) nanowire joining. (e) metallic (Ag) and semiconductor (CuO) nanowire joining. (f) Semiconductor nanowire (TiO_2) on metallic electrodes. (a, b) reproduced with permission from Ref. [144]. Copyright 2015, Elsevier. (c) reproduced with permission from Ref. [147]. Copyright 2015, The Optical Society. (d) reproduced with permission from Ref. [148]. Copyright 2016, IOP Publishing. (e) reproduced with permission from Ref. [138]. Copyright 2020, AIP Publishing. (f) reproduced with permission from Ref. [136]. Copyright 2016, Wiley-VCH.

Joining mechanism for non-metallic nanomaterials

Following photo-absorption leading to electronic and lattice excitation, the relaxation of absorbed energy occurs as described in the previous section. In non-metallic nanomaterials such as semiconductors and insulators, the deposition of absorbed laser energy eventually results in heating of the lattice [164]. As there is no SPR effect for non-metallic nanomaterials, heating arises from the direct excitation of electrons forming electron-hole pairs or excitons that then deposit energy at localized defect sites within the lattice. When exposed to high intensity picosecond or longer duration laser pulses, the excitation of carriers heats the lattice as in photo-thermal processing. With control over spatial localization and the beam thermal profile, laser processing can result in melting of non-metallic nanomaterials while minimizing the heating effect in the surrounding region and in the underlying substrate. Joining occurs when adjacent nanostructures are heated to the melting temperature under laser irradiation and then merging or coalescence to form a bonded structure [182].

When non-metallic nanomaterials are joined using laser irradiation, structural changes, defect generation, stoichiometry changes and doping can occur. This will potentially change the optical, thermal, chemical and physical properties of the non-metallic nanomaterials participating in the joining process which may affect the performance when these materials are incorporated in devices and applications [183]. For example, Pan et al. observed a five-fold increase in the electrical conductivity of ITO nanoparticles after laser processing due to the generation of defects [184]. In another example, Kang et al. found that laser processing of CuO nanoparticles not only induced the joining of these nanoparticles, but also reduced CuO to form metallic Cu nanoparticles. This provided a method for the single-step fabrication of Cu NP based electrodes [185].

When ultrafast laser radiation is used to join non-metallic nanomaterials, a variety of new non-thermal physical mechanisms need to be considered. These arise when intense excitation momentarily transfers the system to a highly non-equilibrium state.

The resulting disordering of the lattice and the transient elimination of crystalline structure in non-metallic nanomaterials is a unique effect that can only be produced under excitation with high intensity ultrashort (fs) laser pulses [164,165]. For example, Xing et al. found that the excitation of excitonic states via two-photon absorption occurs in a ZnO nanowire under femtosecond laser irradiation, creates an electron-hole plasma that generates localized heating as well as defect generation at the junction between ZnO-ZnO nanowires during joining [151]. While these non-equilibrium processes dominate the initial stage of the laser-matter interaction with ultrashort laser pulses, some energy in the system is eventually transferred as heat to the non-metallic nanomaterials producing a photo-thermal effect in the overall system.

Laser-induced joining of nanomaterials

Joining of zero-dimensional nanomaterials

Joining of homogeneous nanoparticles

Joining of metallic nanoparticles

Metal nanoparticles have attracted much interest because these materials have demonstrated to be ideal for a wide variety of practical applications as components in the next generation of electronic and optoelectronic devices and in catalytic systems. The feasibility of joining metallic nanoparticles was first studied in 2005, using a 532 nm 30 picosecond Nd:YAG laser to weld Au nanoparticles on a carbon coated transmission electron microscopy (TEM) grid [143]. It was found that a nanocontact having an ohmic contact can be achieved by intense laser pulse irradiation. A single phase solid with a clear necking structure was formed by thermal heating during laser irradiation (Fig. 6(a), (b)). It was also found that laser irradiation initially causes gold nanoparticles to associate with each other and that further irradiation results in welding of these associated particles. It was proposed that these processes were the result of localized heating arising from photon absorption, but there was no discussion of the effect of plasmonic heating on the joining mechanism. Another early study investigated the characteristic coalescence time for paired Au nanoparticles as observed using in-situ optical extinction spectroscopy under ns laser irradiation [186] and found that a pair of Au nanoparticles, each with 100 nm diameter coalesced in less than one ns. This conclusion was supported by a molecular dynamic simulation.

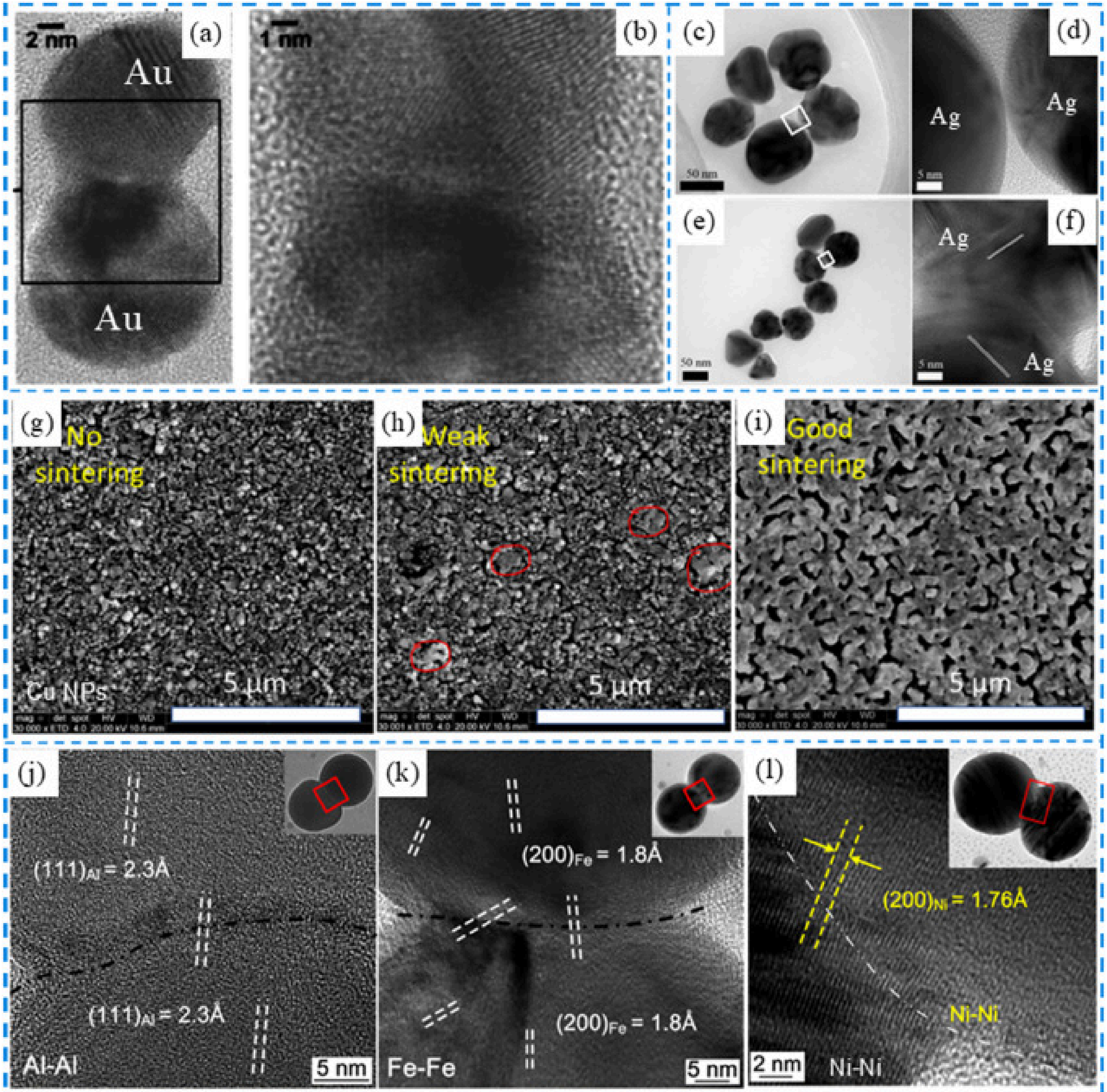

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Fig. 6. Laser-induced joining of metallic nanoparticles. (a, b) Au nanoparticle joining by picosecond laser. Reproduced with permission from Ref. [143]. Copyright 2005, AIP Publishing. (c-f) Ag nanoparticle joining by fs laser irradiation. (c-d), low fluence fs laser irradiation only decomposes the PVP shell on the Ag nanoparticles; (e-f), high fluence fs laser irradiation promotes softening and Ag atom diffusion in the gap forming a metallic necking region bridging the Ag nanoparticles. (c-f) Reproduced with permission from Ref. [159], Copyright 2012, AIP Publishing. (g-i) Morphology of Cu nanoparticles with laser-induced sintering. Reproduced with permission from Ref. [171], Copyright 2018, American Society of Mechanical Engineering. (j) Al nanoparticles joining by fs laser irradiation. Reproduced with permission from Ref. [146]. Copyright 2014, American Chemical Society. (k) Fe nanoparticle joining by fs laser irradiation. Reproduced with permission from Ref. [146]. Copyright 2014, American Chemical Society. (l) Ni nanoparticle joining by fs laser irradiation. Reproduced with permission from Ref. [196]. Copyright 2014, AIP Publishing.

Normally, metallic nanoparticles like Ag nanoparticles are covered with a layer of polyvinylpyrrolidone (PVP) during the chemical synthesis process to prevent agglomeration. This polymer layer is found to influence the joining of Ag nanoparticles in solution. Huang H. et al. has studied this process in the joining behavior of adjacent Ag nanoparticles in response to changes in fs laser fluence [159] (fs laser parameters: 800 nm wavelength, 1000 Hz, 35 fs pulse duration). It was found that bonding between Ag nanoparticles occurs via the formation of an intermediate hydrogenated amorphous carbon layer deriving from the decomposition of PVP at fluences as low as 95 $\mu J/cm^2$ (Fig. 6 (c), (d)) However, at a high fluence of 200 $\mu J/cm^2$, the amorphous carbon layer is replaced with a metallic Ag neck joining the Ag nanoparticles (Fig. 6(e), (f)). This suggests that the joining of Ag nanoparticles by laser irradiation initially proceeds through an amorphous carbon bridge arising from the decomposition of PVP followed by the softening of the lattice at the Ag surface. This creates a metal neck connecting the nanoparticles. Further fs laser irradiation can lead to the coalescence and sintering of additional nanoparticles to produce submicron and micron-sized Ag agglomerates consisting of assemblies of many individual nanoparticles [187], normally suggested as the “sintering” process. The melting and further aggregation of these Ag nanoparticles to larger particles is triggered by localized surface plasmon resonance induced electric field enhancement as “hotspots”. It is also found that the Ag nanoparticles do not need to be bonded prior to irradiation as gaps of up to ~10% of the nanoparticle diameter can be bridged as Ag flows between individual Ag nanoparticles in response to the coupled electric fields induced by plasmons in adjacent structures [143]. The Ag filler materials is from the laser ablated process and redeposited in the bridge structure [188]. It should be noted that the laser power density for the metallic nanoparticles need to be carefully controlled. Irradiation of a Ag nanoparticle solution at high fluence with fs laser pulses has been found to produce photo-fragmentation of Ag nanoparticles and the generation of a large number of tiny Ag nanoparticles [167].

Laser joining/sintering of metallic nanoparticles has attracted much attention as it provides a facile, scalable, controllable and damage-free fabrication process which is compatible with the assembly of nanocomponents on heat-sensitive substrates and in flexible electronics. Laser joining and sintering of nanoparticles is of interest due to the fact that laser-induced photothermal input at selected locations can engineer the local temperature with high selectivity and controllability. Large area joining and sintering can be easily achieved with high laser scanning speed up to 3000 mm/s [153] and short laser exposure time down to 80 µs [189]. A summary of representative research on laser joining metallic nanoparticles, such as Ag [144,145,155,159,166,187,188, [190], [191], [192], [193]], Cu [170,171,194], Au [143,168,195], Al [146], Fe [146] and Ni [196] using different lasers including CW laser [171,191,194,195,197], Nd:YAG millisecond (ms) laser [190], nanosecond (ns) laser [171,198], picosecond (ps) laser [198] and femtosecond (fs) laser [[144], [145], [146],159,166,168,171,187,188,192,196] is given in Table 2. A review about laser sintered metallic nanoparticles can be found in Refs. [199,200]. Another example involving the laser sintering of Cu nanoparticles can be seen in Fig. 6(g-i). It is evident that laser scanning at different speed changes the size range of sintered Cu structures and a suitable processing window can induce large area of joining and further coalescence of Cu nanoparticles into conductive film. A comparison of the conditions for joining/sintering of Cu nanoparticles using different types of laser is also available [171]. It was found that the processing window for high quality sintering is often quite narrow for pulsed laser irradiation, suggesting that control of laser processing parameters is necessary to achieve good sintering characteristics while limiting the heat affected zone. Sintering of Cu nanoparticles with CW laser radiation provides a much larger processing window and gives a more uniform product. However, CW processing is accompanied by a large heat affected zone which may be problematical for applications in flexible electronics.

Table 2. Summary of some joining studies for common metallic nanoparticle using laser processing.

Type of metal/Type of laser	Ag	Cu	Au	Fe	Ni	Al
fs laser	[144,145,159,166,187,188,192]	[171,201]	[168]	[146]	[196]	[146]
ps laser	[198]		[143]			
ns laser	[198]	[171]	[186]			

Type of metal/Type of laser	Ag	Cu	Au	Fe	Ni	Al
ms laser	[190]					
CW laser	[155,191,193,197,198]	[170,171,194,202]	[203]		[202]	

Some examples of laser joining of Al-Al, Fe-Fe and Ni-Ni nanoparticle pairs by fs laser irradiation are shown in the high-resolution transmission electron microscopy (HRTEM) images in Fig. 6 (j) (k) and (l) respectively. These structures are characterized by a necking region with lattice matching as found in the fs laser joining of Au and Ag nanoparticles. These similarities suggest that bonding in all these systems can be attributed to the same physical mechanism: localized heating near the contact region between nanoparticles.

Joining of semiconductor nanoparticles

Semiconductor nanoparticles have been widely used in applications such as photocatalysts, gas sensors, dye-sensitised solar cells, and thermo- and opto-electronic devices. However, the integration of semiconductor nanoparticles in next generation devices remains challenging. In recent years, laser processing of semiconductor nanoparticles has been shown to provide a viable alternative for obtaining joined nanoparticles facilitating the construction of functional thin films with porous, compact morphologies on a variety of substrates.

In an early study, UV laser sintering of TiO_2 nanoparticles was shown to enhance the performance of solar cells due to the removal of organic additives and improvement in the electrical contacts among TiO_2 nanoparticles [204]. A UV laser was used since TiO_2 nanoparticles are highly absorbent at wavelengths less than ~385 nm and the band gap energy of the TiO_2 nanoparticles must be less than the photon energy successful sintering. Similar research on UV laser sintering of TiO_2 nanoparticles has also been carried out under different laser processing conditions [131,[205], [206], [207]] (Fig. 7(a-c)). In general, laser processing provides a method of forming functional units from TiO_2 nano-building blocks on heat sensitive substrates enabling the full engineering of nanoporous TiO_2 structures. Despite this interest, there has been little research into joining mechanisms of nanoparticles under these conditions. In another study, a CW laser with a wavelength of 355 nm was used to join and sinter ZnO nanoparticles (Fig. 7(d-f)) [130]. Densification of the porous ZnO thin film together with coarsening of the particles suggests that the nanoparticles have gone through a molten phase during laser processing. These results suggest that effective heating and consistent sintering are a strong function of the laser wavelength relative to the bandgap of the material.

The initial porosity of the nanoparticle thin film is also key parameters. Studies of the laser-induced aggregation of ZnO nanoparticles using a ns pulsed excimer laser at 248 nm and ps pulses from a 355 nm laser have also been reported [182,208]. Other research on laser-induced joining and sintering of semiconductor Ge nanoparticles [209] (Fig. 7(g-i)), indium-tin-oxide nanoparticles [184,210,211], NiO nanoparticles [212] and PbSe nanocrystals [132] have also been reported.

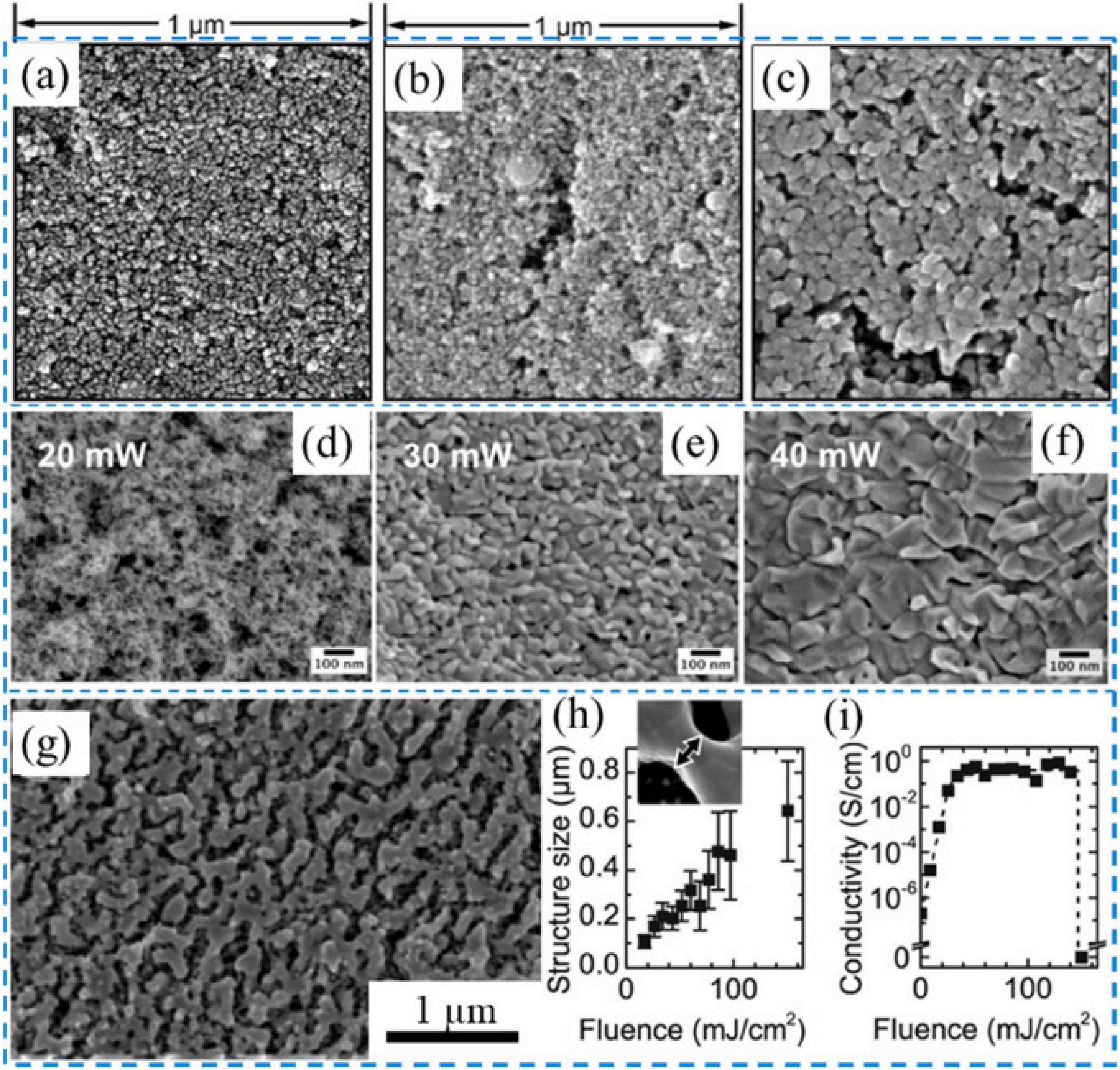

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Fig. 7. Laser-induced joining and sintering of semiconductor nanoparticles. (a-c) TiO_2 nanoparticles. (a) as deposited TiO_2 nanoparticles, (b) morphology TiO_2 nanoparticles after 5 μs laser irradiation at the power density of 0.5 MW/cm^2 with a CW laser with a wavelength of 355 nm, (c) morphology TiO_2 nanoparticles after 50 μs laser irradiation at the power density of 0.5 MW/cm^2. (a-c) Reproduced with permission from Ref. [131]. Copyright 2013, Elsevier. (d-f) ZnO nanoparticles with different laser powers by a CW laser with a wavelength of 325 nm. Reproduced with permission from Ref. [130]. Copyright 2013, AIP Publishing. (g) morphology of sintered Ge nanoparticles with a laser fluence of 17 mJ/cm^2 (laser wavelength: 532 nm, 5-7 ns pulse duration). (h) average size of particle vs. laser fluence. (i) conductivity of Ge nanoparticles thin film vs laser fluence. (g-i) Reproduced with permission from Ref. [209]. Copyright 2013, Wiley-VCH.

It should also be noted that laser-induced reduction of NiO and CuO nanoparticles by laser-induced reduction can produce sintered metallic Ni and Cu nanoparticles for use as electrode [185,213]. This process occurs when solvent molecules or surface ligands act as a reducing agent that converts the metal oxide nanoparticle into a metal under laser irradiation.

Joining of heterogeneous nanoparticles

Joining of heterogeneous metal nanoparticles

Heterogeneous or dissimilar metal joining is attracting much interest in the design of new devices with improved performance and functionality. Even though the joining of different metals has been widely studied on macro- and micro-scales, research on dissimilar joining of nanoscale materials is still very limited. One interesting method is laser-induced melting of one type of metal nanoparticle causing it to bond to another nanoparticle of different composition. This is possible when a plasmon resonance in one type of particle permits it to couple strongly to the incident laser beam. The plasmon energy in the excited particle is then trapped at a hotspot at the interface with the other type of nanoparticle creating a liquid-like state which functions as a “nano-solder” to join the components in the area of contact. This technique for joining was proposed by Mafuné et al. [125] who successfully joined Pt-Au nanoparticles in water with ns pulses from a 532 nm Nd:YAG laser. This wavelength is close to that of the wavelength of the surface plasmon peak (~520 nm) of the Au nanoparticles. Therefore, Au nanoparticles will be heated above the melting point under intense laser irradiation and transformed into liquidlike nanoparticles while Pt nanoparticles remain the solid shape. The interface between Au and Pt is shown in Fig. 8(a-b) where it is apparent that liquid Au flows to form a coating on the surface of adjacent Pt particles that acts to bond the dissimilar nanoparticles. A similar result is obtained when fs laser radiation is used to join Ag and Au nanoparticles [160]. In this case, the Ag nanoparticles are only 8 nm in diameter, while plasmon resonances are generated in both types of nanoparticle. Plasmon heating is then intense in both Ag and Au enabling the Ag nanoparticles to melt and act as a solder (Fig. 8(c, d)). This process is facilitated by a reduction in the melting point of the Ag nanoparticles due to their small size. Since the melting point of the metallic nanoparticles depends to a large extent on the size of the nanoparticles, it would be interesting to study how variation of the size of the Ag nanoparticles in this system affects the joining mechanism. In another report, Semaltianos et al. [162] have demonstrated that picosecond laser irradiation of a mixture of Ag and Pt nanoparticles in solution can lead to the joining of dissimilar nanoparticles and the formation of bimetallic alloys. In a similar experiment, Liu et al. discussed the joining of Pt and Ag nanoparticles by fs laser ablation of a Ag target immersed in an aqueous solution of Pt nanoparticles [161]. It was found that the ablation of the Ag target leads to the appearance of tiny Ag nanoparticles which will have a much lower melting point than that of bulk Pt. These tiny Ag nanoparticles function as nano-fillers and act to join with larger Pt nanoparticles via lattice mismatch at the Ag-AgPt interface (Fig. 8(e, f)).

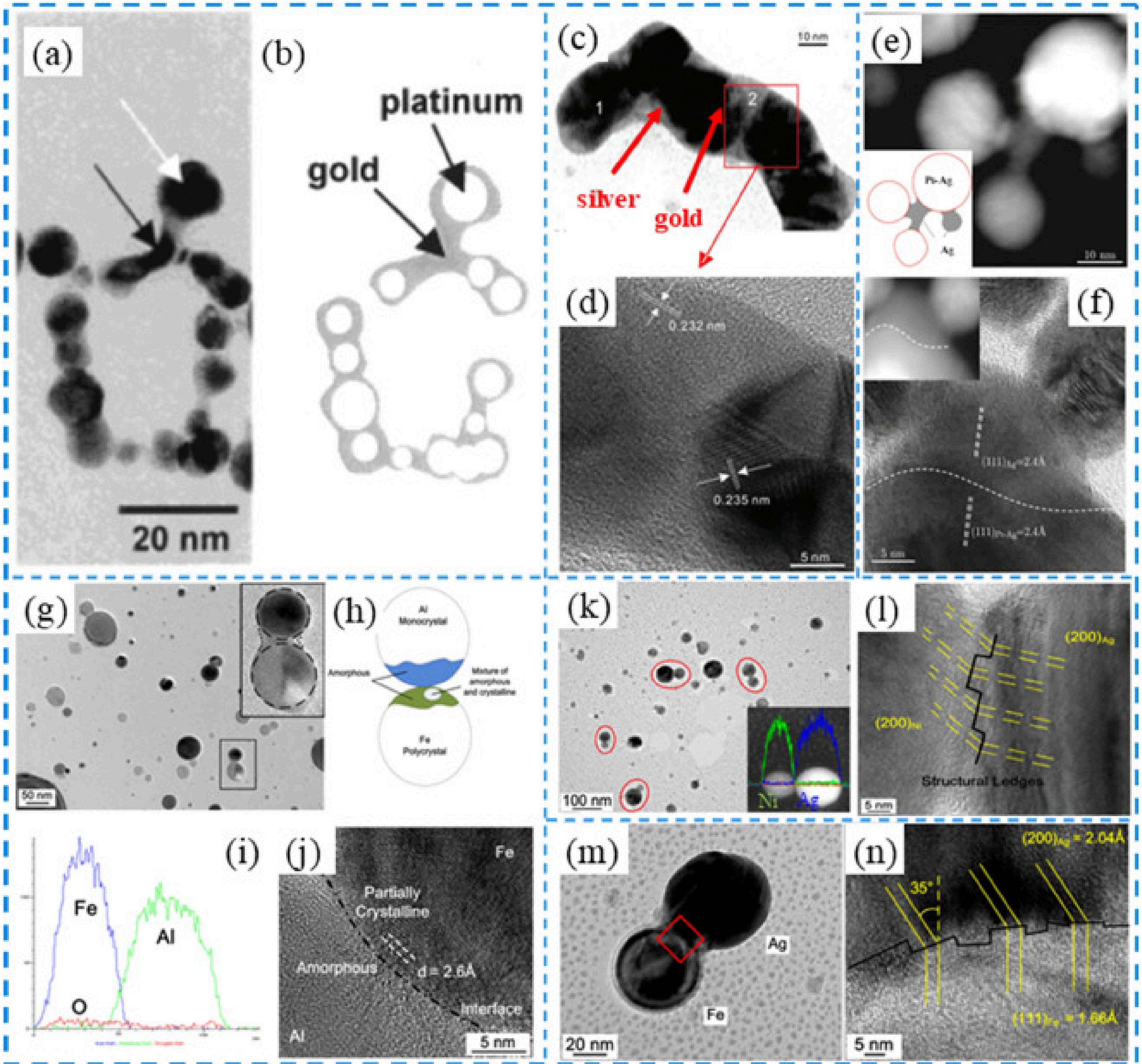

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Fig. 8. Laser-induced joining of dissimilar metallic nanoparticles. (a, b) Au-Pt nanoparticle joining by picosecond laser irradiation. Reproduced with permission from Ref. [125]. Copyright 2003, American Chemical Society. (c, d) Au-Ag nanoparticle joining by picosecond laser irradiation. Reproduced with permission from Ref. [160]. Copyright 2013, American Scientific Publishers. (e, f) Ag-Pt nanoparticle joining by fs laser irradiation Reproduced with permission from Ref. [161]. Copyright 2013, Springer Nature, under Creative Commons Attribution 4.0 International License (http://creativecommons.org/licenses/by/4.0/ ↗). (g-j) Al-Fe nanoparticle joining by fs laser irradiation. Reproduced with permission from Ref. [146]. Copyright, 2014, AIP Publishing. (k, l) Ag-Ni nanoparticle joining by fs laser irradiation. Reproduced with permission from Ref. [196]. Copyright 2014, American Chemical Society. (m, n) Ag-Fe nanoparticle joining by fs laser irradiation. Reproduced with permission from Ref. [196]. Copyright 2014, American Chemical Society.

Jiao et al. have studied the feasibility of joining different metal nanoparticles formed by fs laser ablation of a metallic target in vacuum followed by laser irradiation of the resulting nanoparticles in a vacuum system. It was found that, in addition to the joining of homogeneous nanoparticle pairs such as Ag-Ag, Fe-Fe, Al-Al and Ni-Ni, a fraction of dissimilar nanoparticles was also joined. Such dissimilar combinations included Al-Fe Ag-Fe, and Ag-Ni nanoparticles [146,196]. The lattice structure in the bonded interface of these heterogeneous pairs is of interest because of the appearance of mixed phases. For example, at the interface between Al-Fe nanoparticles [146], the junction as confirmed by energy dispersive X-Ray (EDX) scans, exhibits a mixed amorphous Al and crystalline Fe phase. This suggests that the formation of hotspots accompanied by spallation at these locations can result in atomic intermixing (Fig. 8(g-j)). However, in the case of joining of immiscible Ag-Ni and Al-Fe nanoparticles [196], no mixing of elements was observed at the interface but the structure shows the formation of subclusters of a segregated nanoalloy with shared bonds. Detailed HRTEM characterization of these interfaces suggested that a specific angle between the two matching planes on either side of the interface adjusts to minimize surface energy. Some structural ledges were also formed at the interface to compensate the atomic misfit (Fig. 8(k-n)). These studies have enriched our understanding of the joining mechanism of dissimilar metallic nanostructures and point the way to the implementation of this technique for use in the creation of new compositions for use in a variety of applications in nanodevices and nanosystems.

To reveal the mechanisms associated with structural and compositional changes during laser joining of heterogeneous metallic nanoparticles, Xu et al. studied the joining process of Ag-Au nanoparticles by in-situ characterization using transmission electron microscopy as processing occurred. This was accomplished by introducing a ns laser beam inside the TEM chamber as shown in Fig. 9(a) [214]. Fig. 9(b)-(g) demonstrates the time dependent evolutionary track for joining of Ag-Au nanoparticle under ns laser irradiation (1 ns duration, 25 kHz frequency, 532 nm wavelength). It should be noted that the average laser irradiation power was kept very low (0.4-3 mW) and the duty cycle was carefully selected (10 pulses on followed by 90 pulses off) to permit the laser joining process to be slow enough to be revealed by TEM characterization. This sophisticated technique can be used to follow the mechanisms involved in Ag-Au nanoparticle joining, and the evolution of bonding under laser irradiation can be systematically studied. It was found that, with an increase in laser irradiation time, a neck region was seen to form at the interface between the nanoparticles, as expected from the trapping of plasmonic energy in the region of the joint. The neck region expanded with time revealing the dominant dynamics of the joining process for metallic nanoparticles under laser irradiation. Furthermore, in-situ High-angle annular dark-field (HAADF) imaging in Scanning TEM (STEM) mode (Fig. 9(h-1) to (k-1)) and EDX mapping (Fig. 9(h-2) to (k-2)) of the Ag-Au nanoparticles under laser irradiation demonstrated that joining begins with the formation of a Ag necking region between the two particles due to the plasmonic effect. A silver shell (thickness ~2 nm) then moves to cover the Au nanoparticle forming a core-shell Au@Ag structure. It is evident that this occurs due to the softening of Ag to form a quasi-liquid phase that migrates to cover the adjacent Au nanoparticle. The coating is found to be in the form of an Ag/Au alloy (Fig. 9(j-k)). In-situ TEM characterization with the coupling of laser processing then provides an ideal tool for understanding the mechanism and evolution for dissimilar nanoparticle joining by laser irradiation. Further research using this technology to investigate nanowire-nanowire joining, nanowire-nanoparticle joining, and joining in more complex geometries will give much needed insight into laser-induced nanojoining in general. The role played by different laser parameters on the joining process can also be studied using this technique.

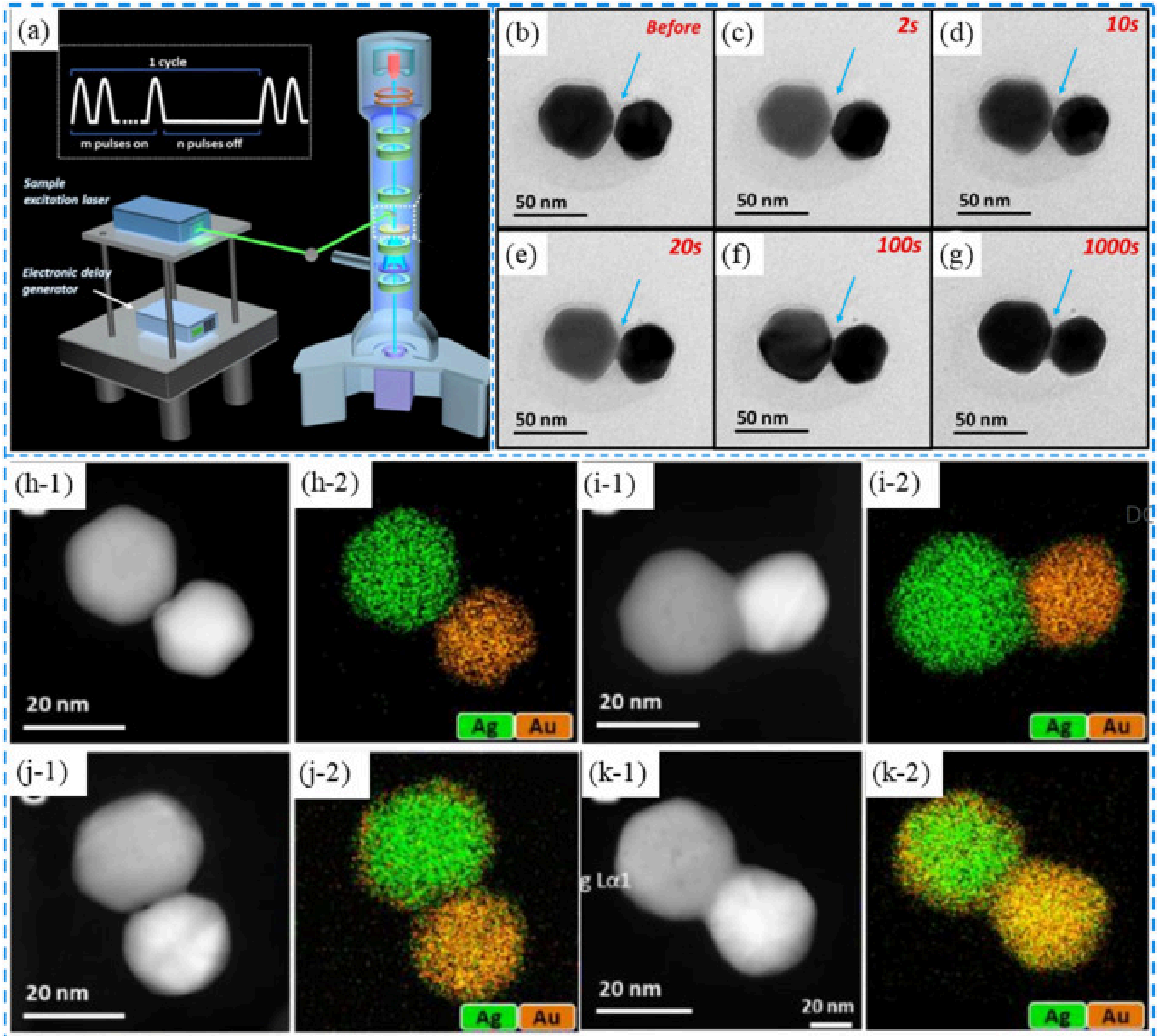

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Fig. 9. In-situ TEM characterization of ns laser-induced joining process in Ag-Au nanoparticles. (a) Experimental set-up. (b)-(g) bright-field TEM images of the Ag-Au nanoparticles demonstrating the evolution under laser irradiation. The arrow indicates the

interfacial region. (h-k) HAADF-STEM images (h-1 to k-1) and EDX mapping (h-2 to k-2) explore four different configurations for the Ag-Au nanoparticle joining under laser irradiation. Reproduced with permission from Ref. [214]. Copyright 2018, Royal Society of Chemistry.

Joining of metal and semiconductor nanoparticles

Metal-semiconductor nanocomposites are particularly interesting for applications in the development of photocatalysts, sensors, solar cells, and biomedicine. As early as 2010, Bajaj and Soni used a Nd:YAG laser with a wavelength of 532 nm to study the joining of Au and SnO_2 nanoparticles synthesized by laser processing in liquid [215]. The Au nanoparticles were found to have a characteristic plasmon resonance peak at 519 nm, enabling direct photon-plasmon coupling. Therefore, during irradiation of the mixed solution of Au and SnO_2 nanoparticles, Au is melted via plasmonic heating and acts as a solder that joins the two types of nanoparticle forming an Au-SnO_2 nanocomposite (Fig. 10(a, b)). This is similar to the soldering behavior of Au-Ag nanoparticles as described in the above section. The morphology of the nanocomposite depends on the laser irradiation time. It was found that irradiation for 5 min leads to the interconnection of Ag and SnO_2 nanoparticles (Fig. 10(a)), while irradiation for 30 min induces a more significant soldering effect that gives rise to elongated particle morphology and an increase in the size of the nanoparticles (Fig. 10(b)). The interfacial structure between the laser processed Ag-SnO_2 nanocomposites was not studied, so no details are available about the nature of the connecting material. Similarly, laser irradiation (1064 nm wavelength, 8 ns pulse duration and 10 Hz frequency) of a mixture of Ag and TiO_2 nanoparticles in solution is also found to lead to the formation of Ag-TiO_2 nanocomposites (Fig. 10(c)), in which a (101) rutile phase matches with the (004) Ag phase at the interface, as shown in Fig. 10(d)) [158]. Other types of metal-semiconductor nanocomposites such as Ag/SnO_2, Ag/ZnO_2, Pt/TiO_2, Pt/SnO_2, Pt/ZnO can be joined using the same approach, providing a simple, highly controllable method for the synthesis of metal-semiconductor nanocomposites as required for the generation of new materials for photocatalytic applications [216].

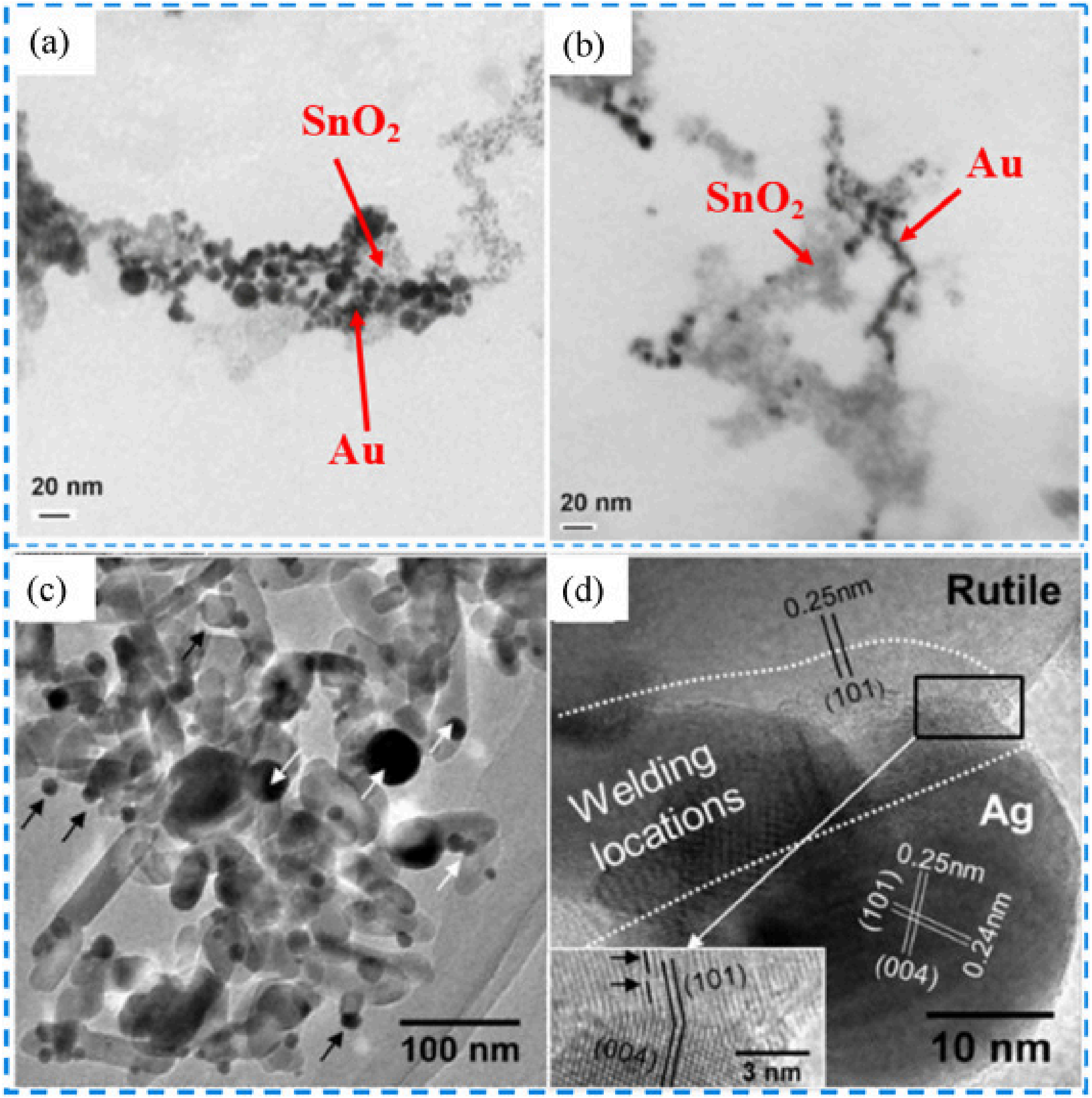

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Fig. 10. Laser-induced joining of metal-semiconductor nanoparticle composites. (a-b) Au-SnO_2 nanoparticle joining. Reproduced with permission from Ref. [215]. Copyright 2010, Springer Nature. (c-d) Ag-TiO_2 nanoparticle joining. Reproduced with permission from Ref. [158]. Copyright 2017, Royal Society of Chemistry.

Joining of one-dimensional nanomaterials

Joining of homogeneous nanowires

Joining of metallic nanowires

Metallic nanowires such as Ag, Au and Cu nanowires have attracted much attention over the last decade because of potential applications as transparent electrodes and as photo catalysts. The percolation of these nanowire networks can be ideal for replacing the expensive ITO conductive electrodes. These nanowire networks are a lost-cost alternative to ITO currently used in conductive electrodes. However, chemically synthesized metallic nanowires often retain a polymer shell such as the PVP layer on Ag nanowire and the cetyltrimethylammonium bromide (CTAB) layer on Au nanowires. Because of this limitation, as prepared metallic nanowire networks normally exhibit poor/unstable electrical conductivity arising from the presence of a large contact resistance due to weak inter-wire bonding and the presence of polymer shells. A large standard deviation in the sheet resistance can represent a big problem for applications such as touch panels, in which the standard deviation must be lower than 10% to precisely locate the touching coordinate. Laser irradiation of metallic nanowire networks is known to yield controllable local joining due to the selective nature of plasmonic induced thermal heating. Use of laser irradiation then provides a promising new technique for scalable and controllable patterning as well as for the fabrication of large area electrodes.

The joining of Ag nanowires via optical welding has been extensively studied and includes the use of a broadband continuous UV light source [77]. It was shown that the electromagnetic field enhancement in the nanoscale gap between two crossed Ag nanowires was sufficient to generate the heat required for joining. However, because the broadband light is absorbed over the entire structure including the underlying substrate, the process is inefficient and is accompanied by the generation of thermal damage in the substrate. These effects are especially serious when the Ag nanowires are supported by a flexible polymer substrate. More selective nanojoining of Ag nanowires is possible when a laser source is used for irradiation. Table 3 summarizes some studies of laser joining of Ag nanowires with CW [149,150,217], ns [129,[217], [218], [219], [220]] and fs laser irradiation [148,[221], [222], [223], [224]] with various of wavelengths from UV to IR range as well as with light from an incoherent source.

Table 3. Summary of reports on laser-induced joining of Ag nanowires. A reference to joining of Ag nanowires with incoherent light is also included for comparison.

Type of laser	Laser parameters (wavelength/pulse duration/frequency)	Power density/spot size in diameter	Methods of Joining	Processing time/number of pulses	Year	Ref.
CW laser	532 nm	120 mW/430 nm (8.26E7 W/cm^2)	Focused	1s	2016	[149]
CW laser	532 nm	200 mW/500 nm (1.02E8 W/cm^2)	Focused	1s	2016	[150]
CW laser	532 nm	(6.3-50) E4 W/ cm^2/10 μm	Scanning	0.1 ms/per spot size	2017	[217]
ns laser	355 nm/3ns	0-65 mJ/cm^2/0.5 mm^2	Scanning	30 μs/1000 pulses	2012	[218]
ns laser	1064 nm/18 ns/150 kHz	15-75 mJ/cm^2/5 μm	Focused	6.7 μs/1pulse	2014	[220]
ns laser	355 nm/27 ns	23.5 mJ/cm^2/500 μm	Scanning	625 pulses	2015	[219]
ns laser	248 nm/25 ns/10 Hz	20 mJ/cm^2/8 × 8 mm	Scanning	0.25-3.75 μs/10-150 pulses	2015	[129]
ns laser	532 nm/5 ns/10 Hz	37 mJ/cm^2/6 × 6 mm	Scanning	5 ns/1 pulse	2017	[217]
fs laser	800 nm/40 fs/1000 Hz	1.5W	Focused	2 ms	2015	[222]
fs laser	800 nm/50 fs/1000 Hz	90 mJ/cm^2/2 mm	Irradiation	~10 s	2016	[148]
fs laser	800 nm/50 fs/1000 Hz	45-200 mJ/cm^2/500 μm	Scanning	0.05-5 s/50-5000 pulses	2016	[221]

Type of laser	Laser parameters (wavelength/pulse duration/frequency)	Power density/spot size in diameter	Methods of Joining	Processing time/number of pulses	Year	Ref.
fs laser	800 nm/120 fs/80 MHz	0.2-2.0 μJ/cm^2/800 μm	Scanning	0.16-1.6 s/1.28 × 10^7-10^8 pulses	2019	[223]
fs laser	800 nm/120 fs/1 kHz	35 mJ/cm^2/170 μm	Scanning	1.7 s/1700 pulses	2019	[224]
Optical welding	Broadband tungsten-halogen lamp	30 W/cm^2	Irradiation	10-120 s	2012	[77]

Dai et al. used a CW laser beam focused on the contact between two Ag nanowires and found that successful joining could only be obtained at certain beam positions [149]. For example, when the laser beam is focussed directly on the contact, one end of the Ag nanowire is melted and shrinks to form a spherical structure (Fig. 11(a)). No joining is possible under these irradiation conditions. However, if the laser beam position is shifted away from the contact point by approximately half the beam diameter, successfully joining was achieved due to heat transfer of the correct amount of energy to the contact (Fig. 11(b)). In this case, the heat generated in the long Ag nanowire is collected in the contact area so that joining occurs under controlled conditions. The efficiency of this process will be dependent on the relative orientation of the long axis of the Ag nanowire and the direction of polarization of the laser beam. The formation of a spherical structure at the end of an irradiated Ag nanowire, followed by migration of this sphere along the nanowire has also been exploited to convert the Ag nanowire into a nanosolder as shown in Fig. 11(c)-(f) [150]. Similarly, Au nanowires can be joined using the same focused laser irradiation [225].

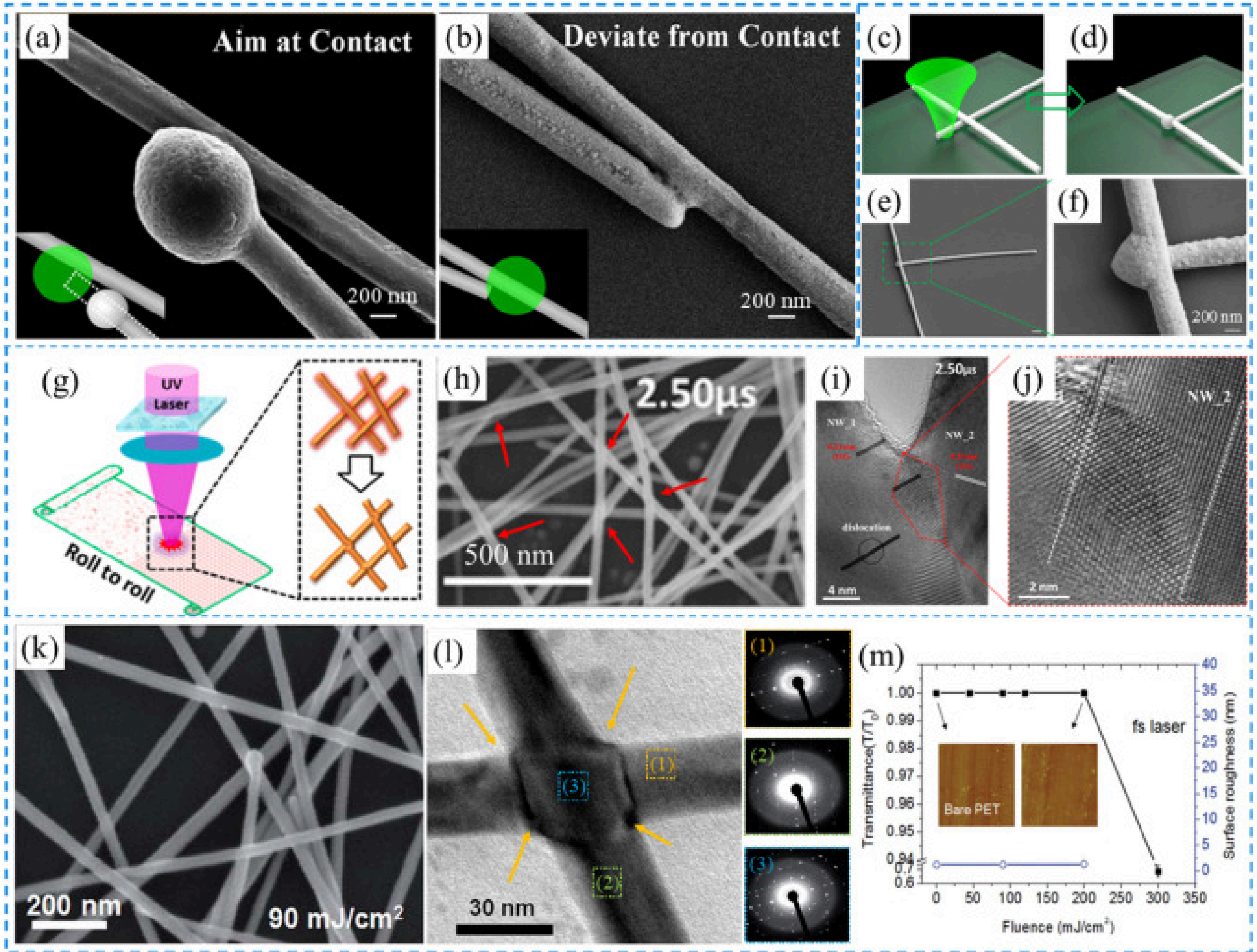

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Fig. 11. Laser-induced joining of Ag nanowires by different types of lasers. (a-f) Focused CW laser. (a, b) Reproduced with permission from Ref. [149]. Copyright 2016, AIP Publishing. (c-f) Reproduced with permission from Ref. [150]. Copyright 2016, AIP Publishing. (g-j) ns laser. Reproduced with permission from Ref. [129]. Copyright 2015. American Chemistry Society. (k-m) fs laser. Reproduced with permission from Ref. [221]. Copyright 2016, Royal Society of Chemistry.

Nian et al. [129] applied a scanning ns laser beam (wavelength 248 nm, 25 ns pulse duration) to locally fuse Ag nanowires and improve the electrical contact at junctions (Fig. 11(g)). It was demonstrated that Ag atoms are strongly heated due to the excitation of plasmons and melt forming a liquid that flows over the nanowire junction before recrystallizing to form the soldered connection. A completed joint is obtained after irradiation for ~ 2.50 μs (Fig. 11(h, i)). It can be seen that the Ag nanowires away from the junction were unaffected by laser irradiation as no changes in overall structure was observed (Fig. 11(j)). This is in contrast to the results obtained using CW laser processing, where a large area is heated and melted. Laser-induced joint formation leads to a significant decrease in the contact resistance of Ag nanowire networks, lowering the total sheet resistance while maintaining high optical transmission. This combination of properties makes pulsed laser processing very promising for applications in the manufacture of screen displays, solar panels, smart windows and other optoelectrical devices.

Irradiation with fs laser pulses produces even less thermal heating than that obtained under ns laser and longer pulse laser irradiation, resulting in efficient joining while avoiding collateral heating in surrounding materials or in the substrate. This can be seen in the fs laser-induced joining of Ag nanowires on a flexible polyethylene terephthalate (PET) substrate [221]. In contrast to the recrystallized joint structure found after ns laser irradiation (Fig. 11(i)) the junction created by fs laser irradiation clearly shows two distinct diffraction patterns belonging to the two Ag nanowires, suggesting that melting and recrystallization does not occur under fs laser irradiation (Fig. 11(k, l)). While a small amount of surface melting can be seen near the point of contact point at the junction, there is little damage to the PET substrate compared to that observed after ns laser irradiation. This effect can be seen in the fluence dependence of surface morphology and optical transmittance of a PET substrate shown in Fig. 11(m). These results indicate that fs laser irradiation improves sheet resistivity in percolated Ag nanowire networks while introducing better uniformity and minimal damage to the underlying polymer substrate [221,223].

Cu nanowires have also been joined using CW [154], μs [226] and fs laser [227,228]. The main advantage of laser-induced joining of Cu nanowires is that the oxidation of Cu nanowires that occurs during the conventional thermal heating process can be avoided. In addition, laser processing is found to lower the contact resistance between nanowires [154]. An interesting innovation is a two-stage nano-recycling process in which Cu and CuO_x nanowires are produced by thermal oxidation and then reduced by laser processing.194 The resulting Cu nanowires are then joined together [229]. This process is initiated under laser irradiation by a reducing agent such as ethylene glycol.

González-Rubio et al. have examined the possibility of joining Au nanorods by femtosecond laser irradiation [177]. They found that at a fluence of 130 $\mu J/cm^2$, the Au nanorods assemble into oligomers in a tip-to-tip pattern with an inter-particle gap of 1.0 ±

0.5 nm. This suggests that monomers and dimers are formed by photothermal decomposition of the organic linkage molecule which then self-assemble to create small Au nanorods (Fig. 12(a) and (b)). It was observed that no structural connection between the Au nanorods was obtained in this condition, but increasing the fluence to 650 μJ/cm^2 yielded welded oligomers. An example of three nanocrystals that have been joined and which have the same crystallographic orientation is given in Fig. 12(c) and 12(d). Melting and joining is due to the large temperature increase at the inter-particle gap caused by the LSPR under fs laser irradiation. This melts the Au nanorod tips, leading to linked Au nanorod chains. Optical absorption spectra for the Au nanorods have also been studied (Fig. 12(e)). Initially, the Au nanorods were randomly dispersed in the solution, and show a characteristic absorption peak at ~600 nm, corresponding to the maximum in the LSPR for the Au nanorod monomer. After self-assembly of Au nanorods induced by organic linkage, the extinction peak at 600 nm decreases, accompanied by an increase of the peak at 700 nm. This is consistent with an increase of concentration of dimers after self-assembly. No significant differences were observed in the assembly process when irradiation was carried out at 13 μJ/cm^2, as this fluence is too small to induce sufficient thermal heating to melt the organic linkages. At an intermediate pulse fluence of 130 μJ/cm^2, the peak at 800 nm is significantly weaker as monomers and dimers of Au nanorods coexist after decomposition of the organic linkages. However, after irradiation at 650 μJ/cm^2, the extinction spectrum showed a new broad band extending from 900 to 1400 nm. This can be attributed to longer Au nanorods created by the joining of individual Au nanorods. CW laser irradiation of a dispersion of Au nanoparticles and Au nanorods in water results in the appearance of Au nanowires having a single crystal structure [230]. In another experiment, Kim et al. used a 30 ps laser to excite the surface plasmon resonance of self-assembled Au nanospheres coated with a CdS shell, leading to the formation of Au@CdS core-shell nanowires [231]. Similarly, Au@SiO_2 core-shell nanowires can be formed by laser-induced joining of Au nanorods with an SiO_2 shell layer [232].

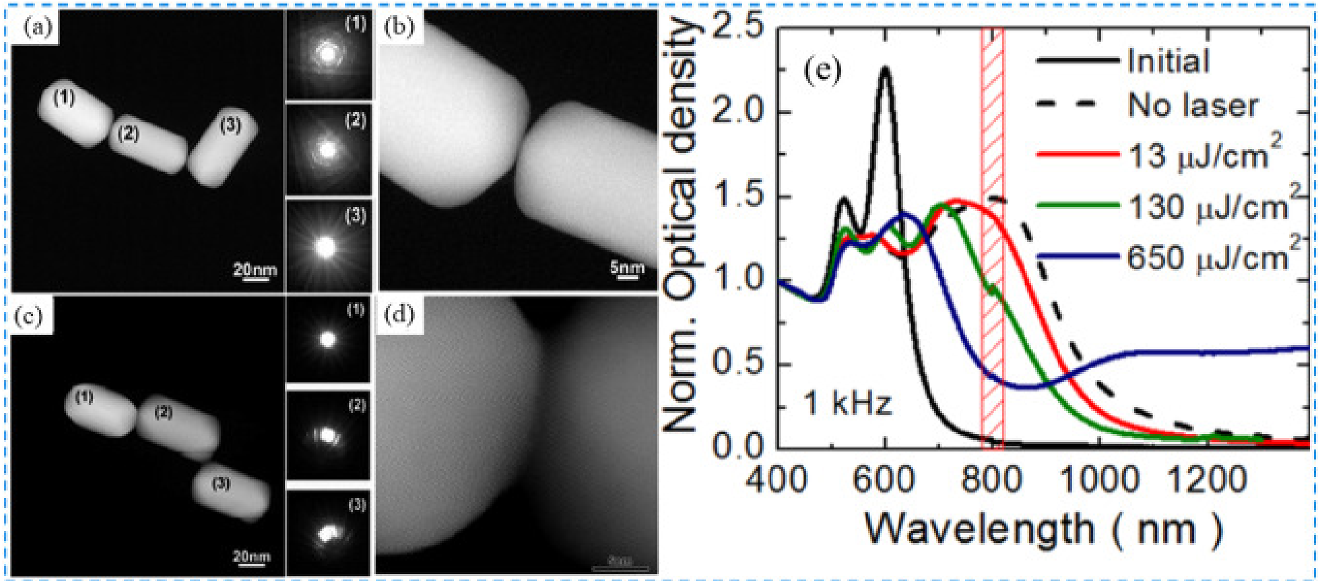

Fig. 12. Assembly and joining of Au nanorods under femtosecond laser irradiation. (a-b) fluence is 130 μJ/cm^2, (c-d) fluence is 650 μJ/cm^2. (e) The optical extinction spectra for the Au-Au nanorod sample after irradiation at different fluences. Initial: before chemical assembly. No laser: after assembly by organic linkers. Reproduced with permission from Ref. [177]. Copyright 2015, American Chemical Society (further permissions related to the material excerpted should be directed to the ACS).

Joining of semiconductor nanowires

Compared to metallic nanowire joining, research on laser joining of semiconductor nanowires is very limited. One of the main reasons for this lack of interest is that plasmonic heating is not available in semiconductors. In addition, most semiconductor nanomaterials have high melting temperatures so that joining by direct heat treatment is not easily implemented. In an early experiment, Yu et al. [233] focussed the output from a He-Ne laser with a wavelength of 632 nm to locally melt the tip of a CuO nanowire forming a joint with an adjacent CuO nanowire (Fig. 13(a)). Similarly, Liu et al. [234] reported the joining of ZnO-ZnO nanowires by focused CW laser radiation at a wavelength of 532 nm, while another study proposed a combination of laser shock peening and laser sintering for joining of CdTe nanowires [235]. During the laser peening process, a shockwave acts to compress the semiconductor nanowire film. This was then followed by ns laser irradiation induced sintering (Fig. 13(b)). This process is quite similar to the laser sintering of metallic and semiconductor nanoparticles as summarized earlier in this manuscript.

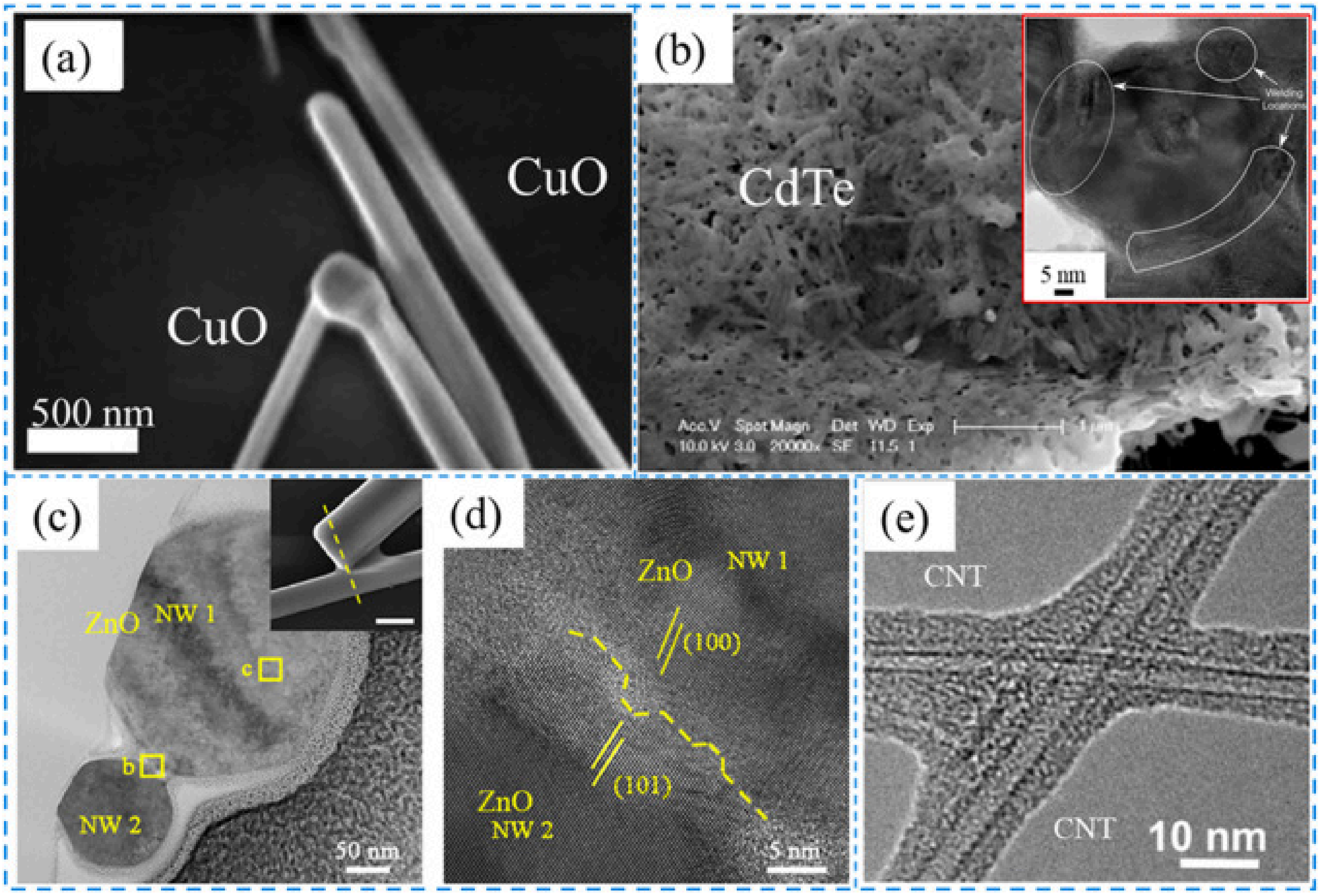

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Fig. 13. Laser-induced joining of semiconductor nanowires. (a) joining of CuO nanowire by He-Ni laser. Reproduced with permission from Ref. [233]. Copyright 2005, IOP Publishing. (b) joining of CdTe nanowire by laser shock peening and laser sintering process, inset shows the HRTEM image of the junction for sintered CdTe nanowires. Reproduced with permission from Ref. [235]. Copyright 2015, Springer Nature. (c-d) joining of ZnO-ZnO nanowire by fs laser irradiation. Reproduced with permission from Ref. [151]. Copyright 2019, AIP Publishing. (e) joining of CNTs by a CO_2 laser. Reproduced with permission from Ref. [236]. Copyright 2007, AIP Publishing.

Xing et al. studied the joining of ZnO-ZnO nanowires with fs laser (Fig. 13(c, d)) [151]. Surprisingly, a solid joint was formed in the junction at the contact between the two nanowires. Xing et al. found that partial melting around the contact area lead to the generation of filler material, which joined the ZnO nanowires forming a homojunction. In fact, the morphology of this junction was found to be quite similar to that occurring due to plasmonic effects in the fs laser-induced joining of Ag nanowires. As no plasmons as such can be generated in ZnO under these irradiation conditions, joining must arise from a different mechanism. A mechanism based on two-photon-absorption was proposed in which the absorption of laser photons gives rise to electron-hole pairs and excitons. These carriers are trapped by lattice defects in the contact region between the ZnO nanowires where they selectively deposit their energy, melting the lattice and facilitating the formation of a joint. This is the first experimental evidence for joining mediated by two-photon absorption in a wide bandgap semiconductor. This work suggests that additional experiments should be carried out for different semiconductor nanowires to see how this process can be used for other joining applications.

Carbon nanotubes (CNT) have received great attention since their discovery in 1991 due to their unique electrical and mechanical properties and their potential applications in electronics, biology, and sensors. A variety of techniques have been used for joining individual CNTs as well as joining CNTs in bulk samples. A review of the joining techniques of CNTs can be found in reference [28]. Early laser-induced bonding of CNTs used a CO2 laser with a wavelength of 10.6 μm to join double-walled CNTs. Bonding was obtained via an amorphous carbon coating on the CNTs after laser irradiation (Fig. 13(e)) [236]. In another study, Yuan et al. examined the joining of CNTs with CW 1064 nm fiber laser irradiation [237]. It was found that the absorption of photons in the CNTs results in the breaking of C-C bonds. The active sites then react to form new chemical bonds in the contact area between CNTs leading to joining. It was also found that the laser irradiation can produce joining between CNTs and a supporting silicon/silica substrate via the creation of Si-C bonds [238].

Joining of heterogeneous nanowires

The joining of dissimilar nanowires provides a direct method for functional engineering of nanowire-based electronic devices. To the best knowledge of the authors, no studies of laser-induced joining of dissimilar metallic nanowires have been reported to date.

Compared with metallic nanowires, semiconductor nanowires provide a wide range of advanced functionalities for use in field effect transistors, P-N junctions, and memory devices. The basic unit for these functional devices takes the form of a semiconductor nanowire in contact with a metallic electrode or a metallic nanowire, where the contact has the properties of a typical metal-semiconductor heterojunction. However, the high surface-to-volume ratio inherent in nanowires implies that the operating characteristics of these devices are strongly dependent on the nature of interfaces between the nanowire and electrodes as well as with other nanoscale materials in the device. This invariably leads to variations in device performance, even for those constructed from the same type of metal-semiconductor combination [14,29,40,41,239,240]. This may limit the future development and application of nanoscale electronic devices incorporating these components. The joining of a metal-semiconductor nanowire heterojunction by laser irradiation provides a simple, non-contact technique for the controlled engineering of the electrical performance of these devices.

Lin et al. pioneered the study of engineering of metal-semiconductor nanowire heterojunction by fs laser irradiation [134]. Joining of Ag and TiO_2 nanowires can be realized with modification of the surfaces of both materials under low fluence fs laser irradiation due to the strong localization of plasmons at the junction. Enhanced wettability of the TiO_2 nanowire surface occurs in the contact region allowing molten Ag atoms to bond to TiO_2 leading to the formation of a robust joint. Detailed studies at the interface of the Ag-TiO_2 heterojunction shows that a highly defective TiO_x layer is formed during laser irradiation, which offers the possibilities of engineering the electronic properties of nanowire devices incorporating these materials (Fig. 14(a, b)). Ghosh et al. applied focused CW laser radiation to irradiate the junction at heterojunction between Ag and ZnO nanowires [135,241]. It was shown that control over the laser power and laser beam position permitted joining of the Ag-ZnO nanowires in different configurations. Junctions could be created by melting of Ag and/or the ZnO nanowire depending on irradiation conditions. An example of joining in this system is shown in Fig. 14(c). Before joining via laser irradiation, the Ag-ZnO junction had low conductivity. After irradiation, and the formation of a robust bond, the heterojunction exhibited rectifying properties that can be attributed to the appearance of a Schottky barrier between Ag and ZnO. However, no additional interfacial structures were found at the Ag-ZnO nanowire heterojunction after irradiation. In a recent study, Xiao et al. further studied the evolution of metal (Ag) and p-type semiconductor (CuO) nanowire heterojunction and the effect on the obtained electrical performance (Fig. 14(d-f) [138]. Exposure to low fluence fs laser pulses for a short period of time results in a structure in which the Ag nanowire remains intact but becomes embedded into the CuO nanowire (Fig. 14(d, e)). This effect occurs as a result of the combination of plasmonic heating in the Ag nanowire and excitonic heating in the CuO nanowire which induces surface melting in the Ag nanowire and softening of the CuO nanowire. In response, the Ag/CuO interface reconstructs to yield lattice matching based on Ag (111)||CuO(111), while both the Ag and CuO nanowires retain their single crystalline structures away from the interfacial region (Fig. 14(f)).

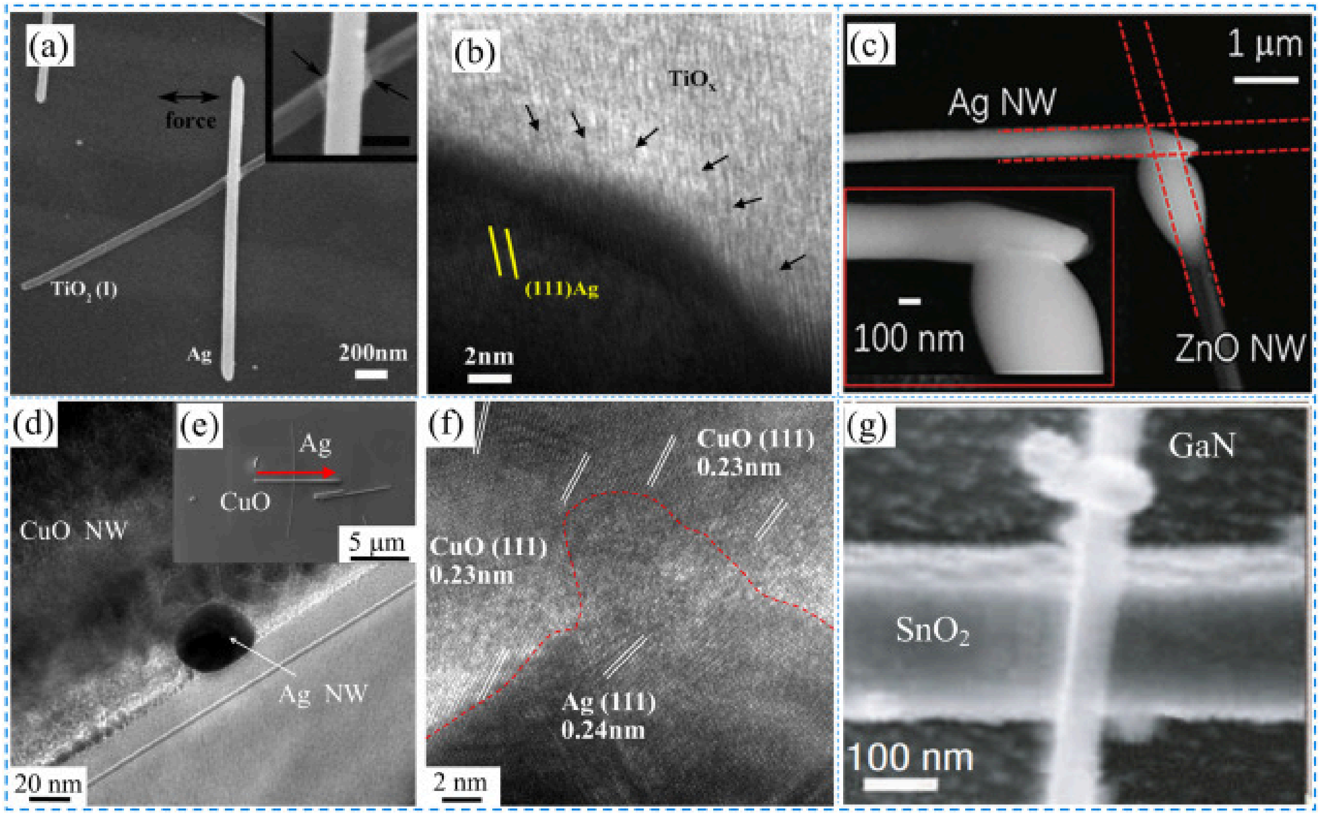

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Fig. 14. Laser-induced joining of heterogeneous nanowires. (a, b) Ag-TiO_2 nanowire joining by fs laser irradiation. Reproduced with permission from Ref. [134]. Copyright 2016, AIP Publishing. (c) Ag-ZnO nanowire joining by CW laser irradiation.

Reproduced with permission from Ref. [135] Copyright 2018, Wiley-VCH. (d, e, f) Ag-CuO nanowire joining by fs laser irradiation. Reproduced with permission from Ref. [138]. Copyright 2020, AIP Publishing. (g) GaN-SnO_2 nanowire joining by CW laser irradiation. Reproduced with permission from Ref. [242]. Copyright 2006, The Nature Publishing Group.

In another report on this subject, Pauzauskie et al. have found that GaN and SnO_2 nanowires can be joined by focusing laser radiation at the point of contact between them after these nanowires located in a water solution with the assist of an optical laser tracking system, as shown in Fig. 14(g) [242]. By carefully controlling the laser power and irradiation time, the GaN and SnO_2 nanowires can be fused together at the junction. However, no further studies were carried out to measure junction properties or identify the lattice structure at the junction interface between these heterogenous semiconductor nanowires.

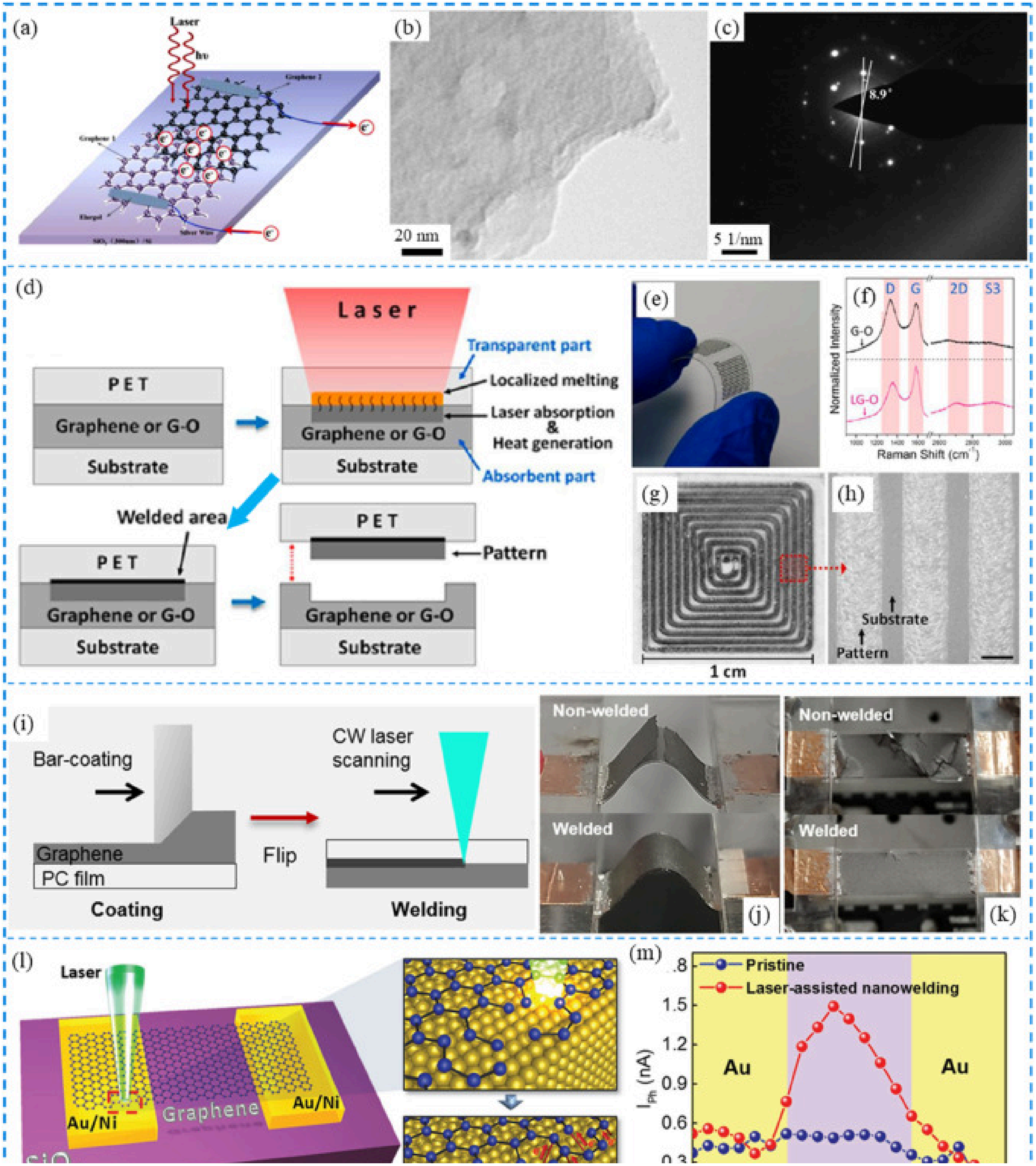
Joining of two-dimensional nanomaterials

There are few reports on the joining of thin films compared to the information available on the joining of zero- and one-dimensional nanomaterials [243]. This is likely due to the fact that the fabrication process for thin films generally involves sputtering; evaporation, or vapor deposition, processes that tend to produce films with good internal bonding and adhesion to substrates. Furthermore, laser irradiation of these films will generate defects as well as damage, which are typically detrimental for thin film-based devices.

Ultrathin two-dimensional materials are now emerging for applications in electronics, environmental systems, and energy generation, so new techniques for joining these 2D materials are highly topical. For example, due to its large surface area and high conductivity, 3D porous graphene is ideally suited for applications in sensors, energy storage, and environmental photocatalysts. Traditional methods for the fabrication of 3D graphene structures including solution processing and chemical vapor deposition yield structures with poor mechanical properties. The high cost and low scalability of these processes are also a concern. One alternative approach is spark sintering of 3D porous graphene oxide flakes which has been shown to result in a welded/joined structure that acts to interconnect a 3D graphene network [244]. These networks, which are promising candidates for medical implants, are highly porous and ultra-low density, while possessing high yield strength and stiffness due to the presence of strong intermolecular bonding between adjacent graphene sheets. The joining of individual graphene sheets to a large area substrate can be beneficial in the construction of microelectronic/optoelectronic devices [245]. Joining of graphene with different concentrations of dopants is a simple way to form P-N or Schottky junctions even in a complex 3D structure [246].

Several additional studies on the joining of 2D materials have been reported using methods such as Joule heating [245], spark sintering [244], ion beam irradiation [246].

As with other materials, laser processing is a simple method for joining of 2D materials such as graphene. In 2013, Ye et al. explored the feasibility of joining two monolayer graphene sheets using a combination of CO_2 laser irradiation and heat from an electrical current [247] (Fig. 15(a-c)). It was shown that joining occurred only by the combined effect of current induced joule heating and laser irradiation. HRTEM images indicate that a low twist angle occurred after the current-assisted laser process, suggesting strong molecular interactions. It was suggested that joule heating from the current source activates the intermolecular attraction (i.e. π-π coupling) while localized laser irradiation provides the energy for further joining of individual graphene flakes.


(a)
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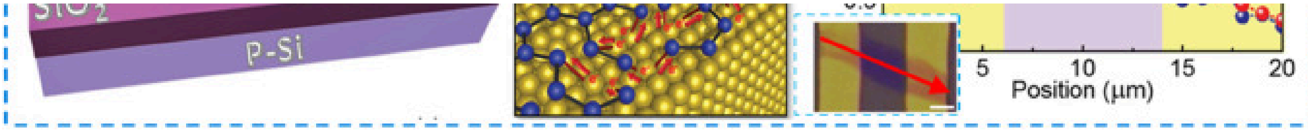

Fig. 15. Laser-induced joining of graphene and graphene with other materials. (a-c) Current assisted CO_2 laser-induced joining of monolayer graphene. (a) schematic of the experimental set-up. (b) HRTEM images of the laser processed bi-layer graphene and (c) selected area electron diffraction (SAED) for the overlapped region. (a-c) Reproduced with permission from Ref. [247]. Copyright 2013, Elsevier. (d-h) laser-induced patterning and transferring of graphene structures. (d) a schematic diagram of this process. (e) image of the laser-induced structure on a PET substrate. (f) Raman characterization of graphene oxide used in the experiment before and after laser processing. (g, h) optical image of the scanned area with different magnifications, scale bar in (h) is 200 μm. (d-h) Reproduced with permission from Ref. [248]. Copyright 2012, American Chemical Society. (i-k) laser-induced joining between graphene and polymer substrate. (i) a schematic of this process. (j) bending test result (k) twisting test result. (i-k) Reproduced with permission from Ref. [249]. Copyright 2019, Elsevier. (l, m) laser-induced welding of graphene on Au electrodes. (l) schematic showing the mechanism for the joining processes, which includes laser-induced structural defects and breaking of C-C bonds and subsequently realization of strong carbon-metal bonding. (m) graphene-based photodetector performance. Inset shows the optical microscope image of the devices and the arrow indicates the direction for laser scanning for the photocurrent performance measurement. (l, m) Reproduced with permission from Ref. [133]. Copyright 2017, Wiley-VCH.

Researchers have also explored the joining of graphene with other materials for selected applications. One of these was described by Oh et al., who proposed a laser transmission welding process that can simultaneously pattern and transfer graphene structures [248]. A CW laser with a wavelength of 976 nm was focused on the graphene or graphene oxide structures placed between two flexible PET thin films. The laser irradiation caused the PET film (transparent at the laser wavelength) to bond with the underlying graphene/graphene oxide component (absorbing at the laser wavelength). The structure after detachment shows the pattern of the graphene structure (Fig. 15(d-h)), even though the width of the pattern is still relatively large (> 200 μm).

Additional focusing to minimize the laser beam size or using a more powerful laser would result in better patterning resolution. In a similar experiment, Choi et al. obtained robust bonding between graphene and a polycarbonate substrate by scanning a CW laser beam [249]. Laser irradiation improved the adhesion and mechanical stability of the joints as measured in tests of bending and twisting (Fig. 15(i-k)). It should be noted that during the laser joining process, the absorption of photon energy on the graphene/graphene oxide will inevitably lead to defect generation in graphene and to the reduction of graphene oxide. In addition, the interfacial structure at the graphene-polymer substrate contact was not well characterized requiring further analysis.

Keramatnejad et al. [133] applied focused CW laser radiation to the interface between graphene and a metal electrode producing bonding of the two structures. The focused laser irradiation was found to selectively dissociate C-C bonds creating chemically active point defects at the edges of graphene within the area of the graphene-metal contact. Graphene-metal bonding at these sites allows carrier transport across the interfacial layer as shown in Fig. 15(l, m) improving the electrical contact. A photodetector constructed from this laser processed graphene-metal junction showed a four-fold increase in photocurrent, suggesting that this type of laser joining process can be used for enhancing the performance improvement of other graphene-based devices. Similar studies of the formation of contacts between graphene and other electrodes by laser-induced thermal effects can be found elsewhere [250,251]. It should be possible to apply selective laser-induced joining of graphene and metal electrodes to other types of 2D structures as part of the engineering of next generation electronic devices.

Joining of nanomaterials with mixed dimensions

Joining of nanoscale materials with different dimensions and morphologies provides a possible route for the fabrication of MEMS and NMES. Combining nanomaterials with different dimensionalities can increase the functionality of new devices for sensing, energy storage, photocatalysts, optoelectronics, and flexible/wearable applications. For example, ultrasonic welding of CNT to a Ti electrode occurs via Ti-C bonds giving rise to a lower contact resistance and improved CNT based transistor performance [48]. Joining of CNT to carbon-coated Si nanoparticles via a hydrothermal process followed by thermal heating produces robust bonding and improvements in electrochemical performance along with long-term stability due to enhanced structural integrity. This combination of properties is necessary for applications as lithium-ion battery anodes [118]. Plasmonic welded CNTs on a monolayer of graphene can also improve sheet resistance while providing additional binding sites for antibody molecules, as required for bio-sensing applications [115]. In another example, Liang et al. demonstrated that flexible graphene oxide layers wrapped and soldered with Ag nanowire networks can lead to lower sheet resistance and high transmittance in the visible range without the need for any heat treatment or laser-induced joining [252]. The existence of graphene oxide ensures that the material has some unique mechanical properties, particularly with respect to stretch-ability. Fabrication of Ag nanowire networks with graphene oxide has resulted in the first fully stretchable polymer light-emitted diodes (LED) devices. The joining of Ag nanowire to graphene, assisted by the creation of Ag-C covalent bonds is also ideal for the production of transparent electrodes with high oxidation resistance and high electrical conductivity [113,116]. In the following section, we will review the status of laser-induced joining of mixed dimensional nanoscale materials, including nanoparticle to nanowire joining, nanowire to two-dimensional/bulk material joining, and nanoparticle to two-dimensional material joining.

Joining of nanoparticles to nanowires

The integration of nanoparticles with nanowires can improve and expand the functionalities of nanowire-based devices. In 2016, Lin et al. [147] studied the assembly of Ag nanoparticles on a Ag nanowire on a silicon wafer by fs laser irradiation. It was found that the electric field enhancement at the Ag nanowire-Si interface under fs laser irradiation attracts adjacent Ag nanoparticles which then self-assemble periodically on the surface of a Ag nanowire when the laser polarization direction is oriented parallel to the long axis of the Ag nanowire, as shown in Fig. 16(b). This type of assembly via optical excitation can be applied in general for the gathering of nanoparticles near nanowires with plasmonic resonances. As an example, Nan and Yan have reported a study of the assembly of Au, Ag and polystyrene (dielectric) nanoparticles on a single Ag nanowire by laser irradiation in water solution [42]. It was found that the Ag nanowire functions as a plasmonic antenna which shapes the electric field in the incident laser beam and guides the optical assembly of nanoparticles. A variety of stable nanoparticle structures can be bonded to the Ag nanowire due to this optical trapping effect. One example of a hexagonal structure of Ag atoms is shown in Fig. 16(a, c, d). This effect can be potentially used for the construction of reconfigurable photonic devices with nanoscale building blocks.

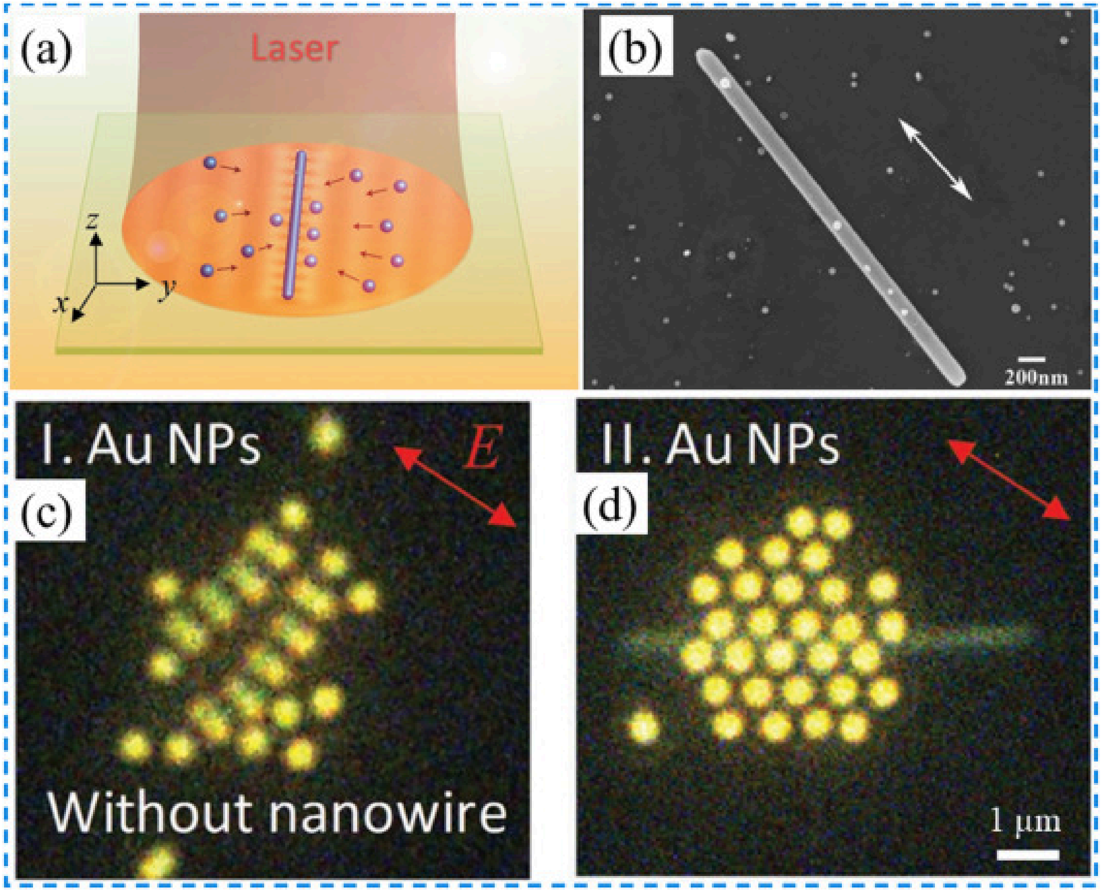

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Fig. 16. Laser irradiation induced optical trapping and binding of metallic nanoparticles on metallic nanowires. (a) schematic diagram of the experimental set up. Reproduced with permission from Ref. [42]. Copyright 2019, Wiley-VCH. (a) periodical assembly of Ag nanoparticle on Ag nanowire under fs laser irradiation. Reproduced with permission from Ref. [147]. Copyright 2015, The Optical Society. The experiment is carried out on a silicon wafer substrate. Laser irradiation induced assembly of Au nanoparticles without (c) and with (d) Ag nanowire. Irradiation is carried out in water solution. Reproduced with permission from Ref. [42]. Copyright 2019, Wiley-VCH.

Metal nanoparticle-semiconductor nanowire hybrid structures are of fundamental interest since they can exhibit the physical properties of both types of nanomaterial. Metal nanoparticles can provide active plasmonic sites that tune the way in which incident light can couple into the semiconductor nanowire, optimizing the overall optical and/or electrical performance of the hybrid structure. A review of metal nanoparticle-semiconductor nanowire hybrid structures and their applications can be found in elsewhere [253]. These studies indicate that the joining of metallic nanoparticles to semiconductor nanowires is of great interest and is worthy of further research.

Joining of nanowires to 2D/bulk materials

Joining between one-dimensional nanomaterials with ultrathin two-dimensional materials such as graphene has attracting much interest. Some studies include the joining of CNTs and graphene using light from a halogen lamp [115], joining a Ag nanowire to graphene by irradiation with an electron beam [113] or from flash light [116]. There are currently no reports on the use of laser irradiation for this application.

Another interesting project involving nanojoining of structures with mixed dimensionality is the use of laser processing to generate an electrical connection between a nanowire and an electrode for use in functional electric devices. It is now apparent that, with the emergence of nanoscale materials as building blocks for functional device units, existing mechanistic models describing the electrical contact between nanoscale materials and an electrode are often not applicable due to the significant difference between the properties of bulk and nanoscale materials [13]. Previous techniques for contact engineering between nanowire and metallic electrodes to improve performance include sonication bonding [48], hot pressing [49], joule heating [254,255], and a combination of localized Joule heating and pressing [29]. Laser processing provides a localized, non-contact method for selective engineering the nanowire-electrode interface, suggesting that it may be a promising alternative to these other technologies for obtaining reliable and controllable performance for nanoscale material-based devices.

The use of fs laser irradiation in this type of application was first identified in 2011 with the publication of a study on local engineering of the interface between an electrode and an In_2O_3 nanowire [30,173]. The notable improvement in the performance of a nanowire-based transistor device following fs irradiation is shown in Fig. 17(a-c). As fs laser irradiation can be accurately directed to specific locations on the contact area, the interface can be selectively annealed without producing significant heating elsewhere in the device. Enhancement in device performance was seen in improved saturation of the source-drain current and a shift in the threshold voltage. The reduction in noise following irradiation can be attributed to an improved contact between the electrode and the nanowire, a reduction in the concentration of interfacial traps and a corresponding reduction in the Schottky barrier height at the contacts. Unfortunately, no structural data is available to correlate interface structure/composition with these improvements in electrical performance.

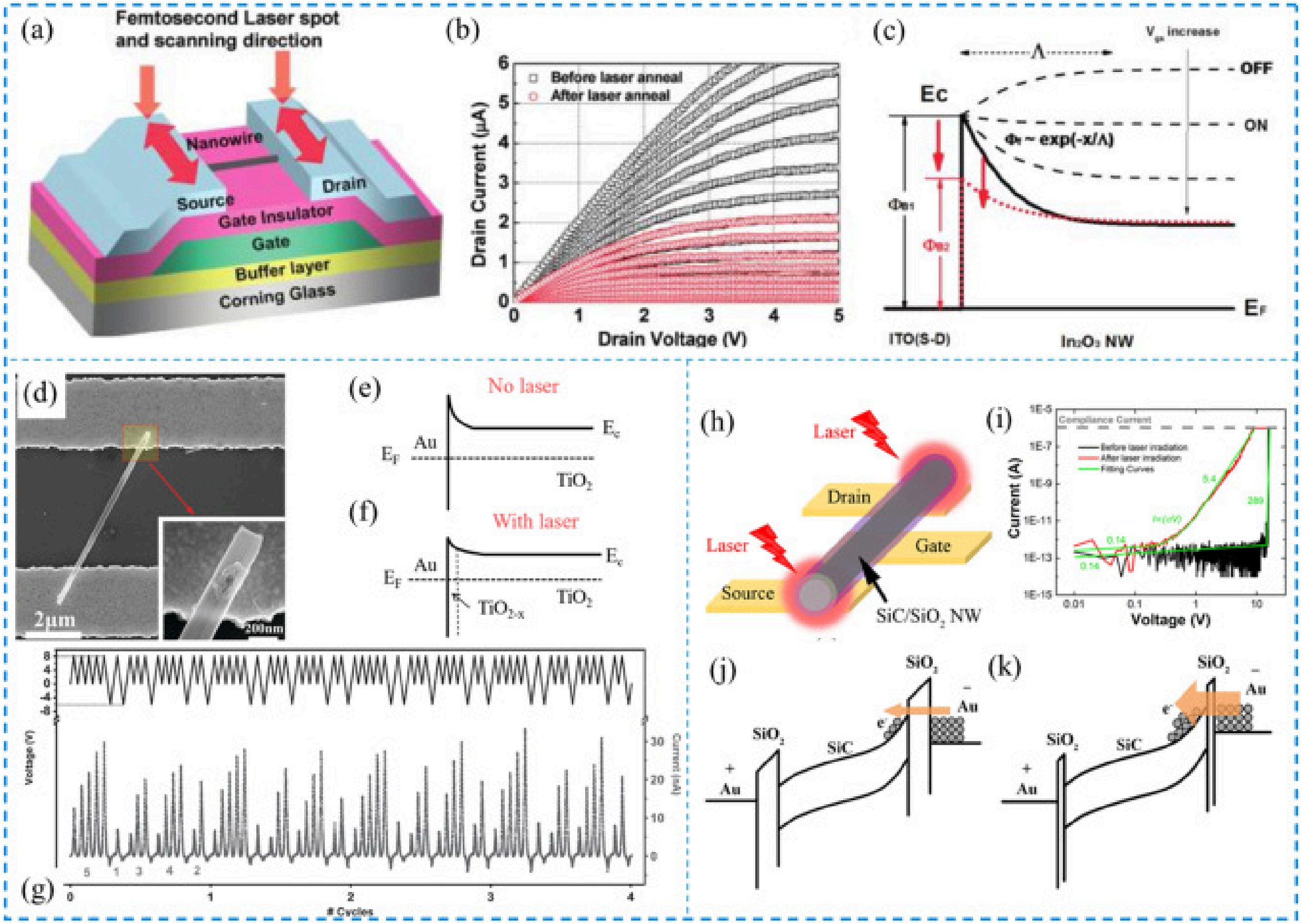

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Fig. 17. Laser-induced contact engineering of a nanowire connected to a metal electrode. (a-c) In_2O_3 nanowire-based transistor devices. (a) schematic of the experimental set-up. (b) the I_{ds}-V_{ds} characteristic curve of an In_2O_3 nanowire transistor device before and after laser engineering. (a) and (b) Reproduced with permission from Ref. [30]. Copyright 2011, American Chemical Society. (c) schematic band diagram between the ITO electrode and the In_2O_3 nanowire before and after laser engineering.

Reproduced with permission from Ref. [173]. Copyright 2011, American Chemical Society. (d-g) fs laser engineered TiO_2 nanowire on an Au electrode. (d) SEM image of the TiO_2 nanowire device. Schematic image for the band diagram before (e) and after (f) fs laser irradiation. (g) Stable multilevel memory performance for a single TiO_2 nanowire device after laser irradiation. (d, g) Reproduced with permission from Ref. [136]. Copyright 2016, Wiley-VCH. (h-k) fs laser engineered SiC/SiO_2 core-shell nanowire on an Au electrode. (h) schematic showing the experimental configuration. (i) electrical performance for SiC/SiO_2 core-shell nanowire-based transistor performance before and after laser irradiation. Schematic image for the band diagram before (j) and after (k) fs laser irradiation. (h-k) Reproduced with permission from Ref. [137]. Copyright 2019, American Chemical Society.

Plasmonic induced heating has also been shown to lead to improved bonding between an electrode and a nanowire. Some nanowire-electrode systems that have been investigated include TiO_2 nanowire [136,256], SiC/SiO_2 core-shell nanowire [137,257], CNT [172] and Cu nanowire [227]. Robust metallurgical bonding (much stronger than a van der Waals bond), between an Au electrode and a TiO_2 nanowire can be obtained after fs laser irradiation. This is accompanied by partial localized reduction of the TiO_2 nanowire in response to plasmonic heating which changes the Au-TiO_2 contact to Au-TiO_x-TiO_2, as shown in Fig. 17(d, e). This engineering of the interface between Au and the TiO_2 nanowire leads to a stable multilevel memory performance such as complex repeatable memory code as demonstrated in Fig. 17(f). Xing et al. further studied the effect of fs laser irradiation on a SiC/SiO_2 core-shell nanowire connected to an Au electrode [137]. It was found that the incident irradiation energy becomes highly localized at the Au-SiO_2 shell interface due to the strong plasmonic effect leading to controllable thinning of the SiO_2 shell layer. The resulting Au-SiO_2-SiC metal-oxide-semiconductor (MOS) heterojunction device was found to have significantly improved performance after fs irradiation. The threshold voltage for switching on was found to decrease from 15.7 V to 1 V, which suggests that charge transfer of carriers between the electrode and the SiO_2 insulator is facilitated after thinning of this layer by fs laser irradiation (Fig. 17(h-k)). Improvements in bonding between Au and the SiO_2 layer after fs laser irradiation was also found to be important in stabilization of the memory application [257].

Joining of nanoparticles to 2D materials

Nanoparticle-2D materials composites such as Ag nanoparticle-graphene nanosheets are of great interest as photocatalysts, for photocurrent generation, in antibacterial applications, and for water treatment due to their unique combination of the optical properties of metallic nanoparticles and the large surface area in 2D materials [[258], [259], [260], [261]]. The strong plasmon resonance of the added metallic nanoparticles can significantly enhances the interaction between light and 2D materials, rendering great promise for the optoelectronic applications of the 2D materials [[262], [263], [264]]. Most traditional methods for the synthesis of nanoparticle-2D material nanocomposites involve chemical absorption or reduction. Laser-induced assembly of composite nanoparticle-2D materials is facilitated by the strong interaction of laser light with nanoparticles. For example, by mixing a solution of Au nanoparticles with a variety of different shapes with graphene oxide followed by irradiation with a nanosecond laser, Zedan et al. [43] demonstrated that the coupling of the surface plasmon resonance in Au nanoparticles to a graphene oxide sheet resulted in the reduction of graphene oxide and the formation of ultra-small (2-4 nm diameter) Au nanoparticles anchored to the graphene oxide surface. The laser irradiation of the Au nanoparticle solution without the presence of graphene oxide only leads to a small reduction in the size of the Au nanoparticles (Fig. 18(a-c)). In the mixed solution, laser irradiation induced defects on the surface of graphene oxide sheets that act as traps for the ultra-small Au nanoparticles bonding them to the surface. Nanocomposites, consisting of ultra-small Au nanoparticles bonded to graphene oxide by laser irradiation, should have many uses as next-generation photo-thermal energy converters for application in novel thermo-chemical and thermo-mechanical devices.

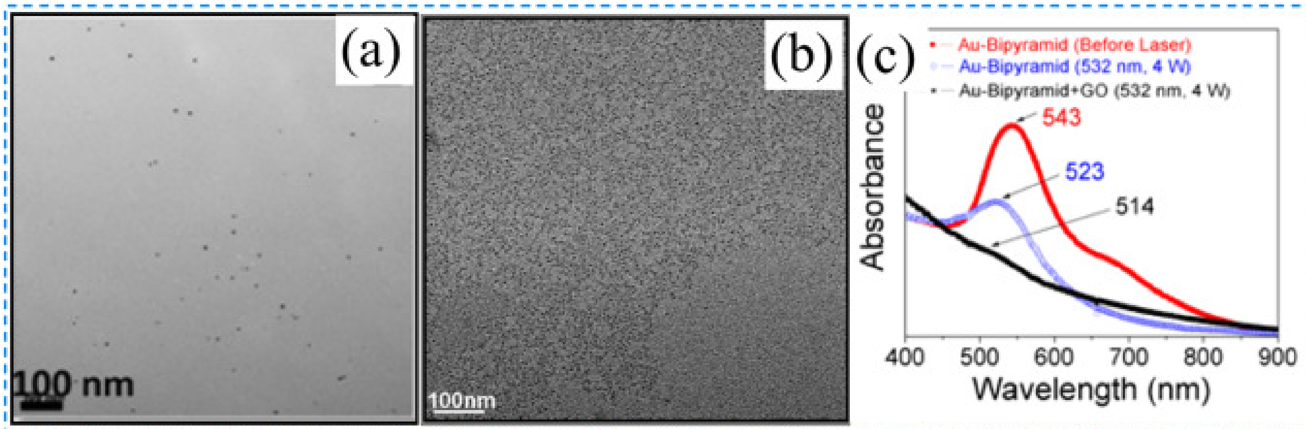

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Fig. 18. Laser-induced assembly of gold nanoparticle on graphene oxide. Laser irradiation of gold nanoparticles in the absence (a) and in the presence (b) of graphene oxide and (c) UV-Vis absorption spectra of the Au nanoparticles after laser irradiation in water (blue curve) and in GO (black curve). Reproduced with permission from Ref. [43]. Copyright 2013, American Chemical Society.

Applications for laser nanojoining

Laser nano-joining provides a straightforward, but fundamental, way to realize permanent bonding between nanoscale building blocks in the assembly of a variety of nanodevices. The attractive features of laser joining, especially for ultrafast laser joining, include a highly localized thermal effect without much heat transfer to surrounding areas such as a flexible substrate. Laser processing gives high spatial selectivity, rapid control over processing parameters and a relatively small environment footprint as required in the integration of diverse passive and active components on flexible or paper substrates [153]. Applications for this technology have been identified in a variety of fields including wearable electronics, robotics, optical devices, energy storage applications, self-powered energy harvesters, bioelectronics and functional textiles. In many ways laser processing technology for nanojoining has already been successfully integrated with ink-jet, screen-printing, and roll-to-roll systems for large area or selective patterning of fabricated electrodes. [265] The following section summaries the recently research for the applications of laser nanojoining: electrode applications, functional nanodevices and optical applications, as shown in Fig. 19.

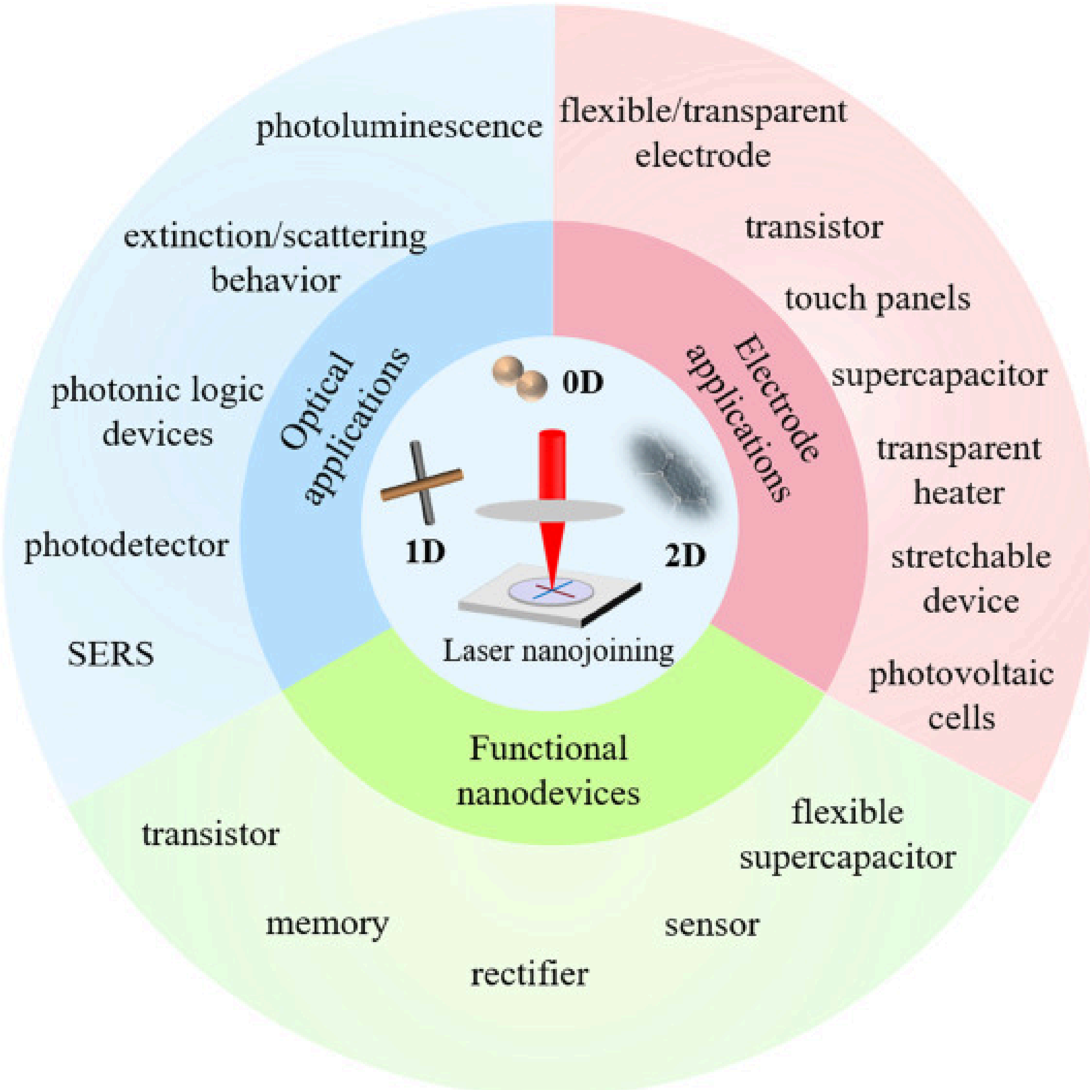

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Fig. 19. Summary of applications for laser-induced nanojoining.

Electrode application

Laser irradiation of metallic nanoparticle based thin films provides a non-contact, damage-free method for the fabrication of large-area fabrication [197]. Furthermore, nanoparticle thin films can be easily coated on different types of substrate by processes such as inkjet printing [265], spin coating, Meyer rod or blade coating [170], screen-printing [266], roll-to-roll printing [193]. Furthermore, by taking advantage of laser processing parameters, patterning of the electrodes can be easily fabricated, providing a direct, simple, repeatable, and scalable method for selective electrode fabrication without the use of conventional vacuum deposition, mask and photolithography processes (Fig. 20(a)). Strong adhesion between the sintered metallic nanoparticle thin film and the substrate can also be achieved [192]. Careful control over laser wavelength, laser pulse duration (to minimize heat diffusion), and the numerical aperture of the focusing lens have been shown to improve resolution of laser sintering. For example, Son et al. have reported the patterning of Ag nanoparticle ink using a 780 nm femtosecond laser with a 100× oil immersion objective lens for the fabrication of a digital nanoscale device [156] (Fig. 20(b)). Uniform metal lines with a minimum width of 380 nm can be obtained. Uniform metal lines with a minimum width of 380 nm can be obtained. These metal electrodes can be further used for the fabrication of functional circuits such as organic transistor [152,156,265] (Fig. 20(c)), strain gauge detection [202], near field communication tag [153], touch panel [170,194,213] (Fig. 20(d)) and supercapacitor [193] (Fig. 20(e)).

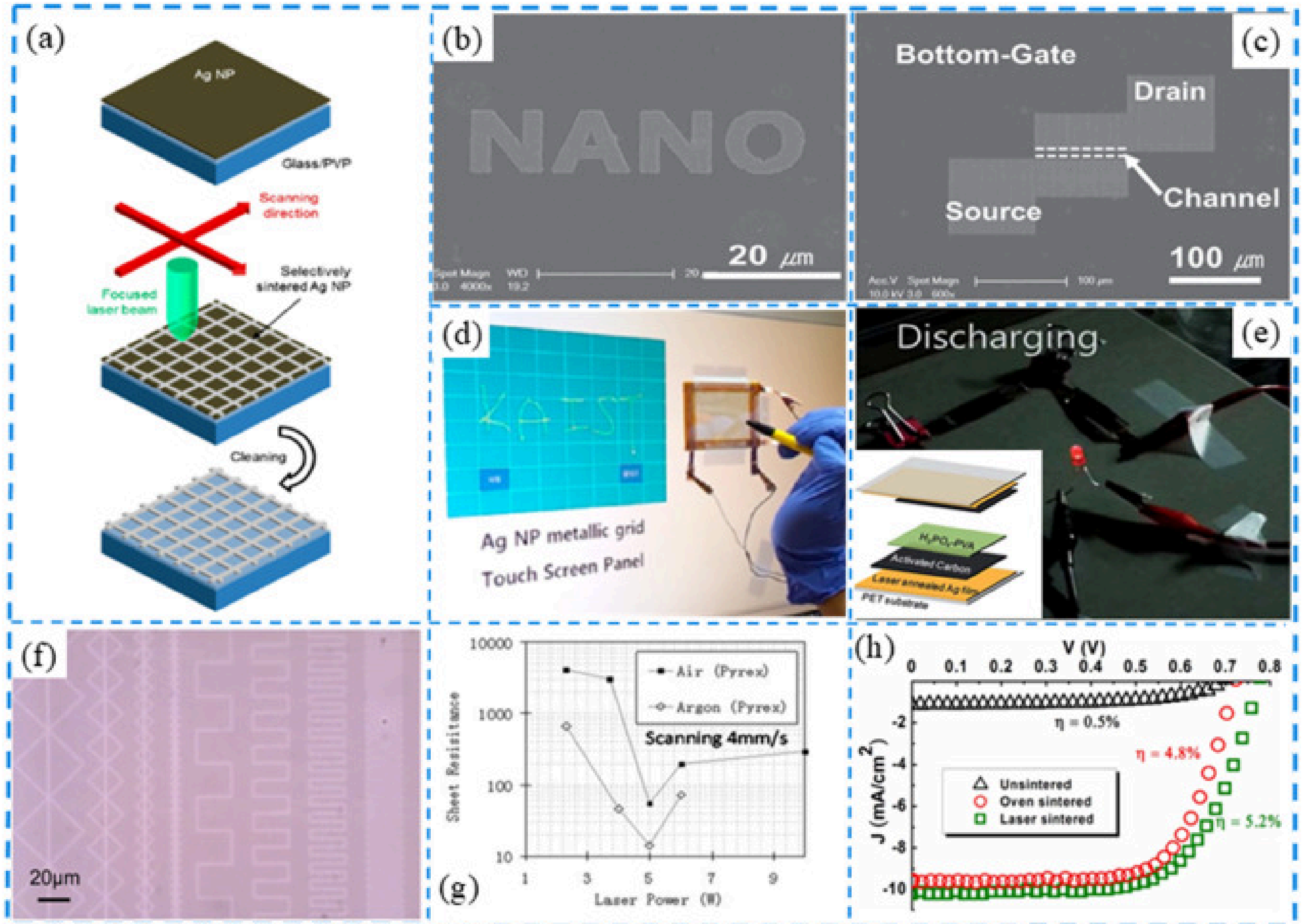

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Fig. 20. Applications of laser-induced joining and sintering of nanoparticles as electrodes. (a) schematic diagram of selective laser sintering and patterning of Ag nanoparticles. Reproduced with permission from Ref. [155]. Copyright 2013, American Chemical Society. (b) SEM image of laser-induced patterning of a Ag nanoparticle thin film showing the letters “NANO”. Reproduced with permission from Ref. [156]. Copyright 2011, Wiley-VCH. (c) SEM image of laser-induced patterning of Ag nanoparticle thin film for fabrication of transistor devices. Reproduced with permission from Ref. [156]. Copyright 2011, Wiley-VCH. (d) demonstration of an application using a transparent metallic grid conductor for a touch panel based on laser-induced patterning of a Ag nanoparticle thin film. Reproduced with permission from Ref. [155]. Copyright 2013, American Chemical Society. (e) lighting of

an LED by a supercapacitor based on thin film of laser sintered Ag nanoparticles, left bottom inset, schematic for the layout of the supercapacitor device. Reproduced with permission from Ref. [193]. Copyright 2015, Royal Society of Chemistry. (f) optical microscope image of 1.5 μm wide line patterns by laser-induced patterning of a ZnO nanoparticle thin film on a soda-lime glass substrate. Reproduced with permission from Ref. [208]. Copyright 2012, Springer. (g) The dependence of sheet resistance on laser power for a laser-induced sintered ITO nanoparticle thin film with a thickness of 900 nm. Reproduced with permission from Ref. [184], Copyright 2011, Springer. (h) comparison of *J-V* characteristics for a TiO_2 nanoparticle based DSSC with different sintering conditions. A conversion efficiency of 5.2% was obtained after laser sintering. Reproduced with permission from Ref. [205]. Copyright 2009, AIP Publishing.

Metal oxide nanoparticles can also be used in a similar way to facilitate joining and controlled sintering. Selective laser processing can generate a patterned conductive metal oxide thin film on a variety of substrates. For example, Lee et al. [208] have reported that focused laser beam processing of a ZnO nanoparticle thin film can print an arbitrary pattern with linewidth as small as 1.5 μm (Fig. 20(f)). The electrical conductivity of a ZnO nanoparticle film can be greatly improved by laser irradiation. This effect occurs because of the generation of oxygen vacancies during the laser irradiation process. This processed material may be useful in the construction of transistor devices [182,208]. Further work has been reported by Pan et al. [184] who demonstrated that laser processing of an indium tin oxide (ITO) nanoparticle thin film with a thickness of 900 nm in argon gas can achieve a sheet resistance of ~14 Ω/sq, which is 4-5 times better than that of commercially available sputtered ITO thin film (Fig. 20(g)).

Metal oxide nanoparticles such as TiO_2 and ZnO are commonly used in dye-sensitized solar cells (DSSC) as a photo-anode. The conventional method for the fabrication of DSSC involves sintering the coated metal oxide nanoparticles at ~450 °C in an oven to remove organic additives as well as to form electrical connections in a nanoporous network within the electrodes. Kim et al. first replaced the traditional process for fabrication of DSSC with laser-induced joining and sintering using a 355 nm CW laser for the fabrication of DSSC [204]. It was shown that the overall conversion efficiency was doubled compared to that of the devices with non-laser treated TiO_2 electrodes. Laser sintering not only yielded good electromechanical bonding between nanoparticles by maximizing the electron diffusion length, but also enhanced the surface area to maximize dye sensitization and light harvesting. Laser processing was found to introduce a neck that acted to join TiO_2 nanoparticles, further improving the electron diffusion coefficient [205,206]. By controlling laser processing parameters such as fluence and scanning speed, a higher conversion efficiency than possible with the traditional oven sintering method was achieved [205] (Fig. 20(h)). This method is also effective in the selective laser sintering of TiO_2 nanoparticles to create an anode in DSSC fabricated on flexible substrates [267].

Similarly, laser irradiation of metallic Ag, Cu, Au nanowire networks (Fig. 21(a)) provides a good alternative for the preparation of transparent electrodes to replace costly ITO thin films. Percolating networks of these metallic nanowires can efficiently conduct electricity while transmitting optical radiation with little absorption. This combination of properties makes these materials promising for use as large area transparent conducting electrodes integrated with the roll-to-roll printing process (Fig. 21(b)) [129]. Laser processing causes localized joining of nanowires at a junction, leading to low sheet resistance and high transmittance in flexible, transparent electrodes with high electrical conductivity. Strong adhesion between the metallic nanowire network and a flexible substrate can also be realized [223]. For example, the sheet resistance of an Ag nanowire network on a PET substrate decreased from 150 to 5 Ω/sq while maintaining an optical transmittance as high as 91% at 550 nm [129], This performance is superior to that of any other commercial flexible ITO thin film in the market (<10 Ω/sq). Using this type of transparent electrode, high-performance touch panels and conductive transparent heaters (Fig. 21(c)) can be realized. Song et al. further demonstrated that laser-induced joining of metallic nanowire nanowires as low sheet-resistance electrodes can also be used for the fabrication of a high-performance solution-processed flexible transistor based on MoS_2 ultrathin films (Fig. 21(d)) [268]. Laser processing of nanowires networks can further lead to improved performance in nanoscale devices for applications in organic photovoltaic applications [219] and stretchable conductor [154]. For example, Spechler et al. demonstrated that laser processed Ag nanowire networks can lead to up to a 33% increase in power conversion efficiency for as fabricated hybrid silicon/organic heterojunction photovoltaic devices [219], as shown in Fig. 21(e). It is evident that laser joining should also be considered as a viable alternative to conventional preparation methods in the assembly of devices incorporating metallic nanowire networks. These would include use in self-powered electrical generator [94] and wearable devices [75].

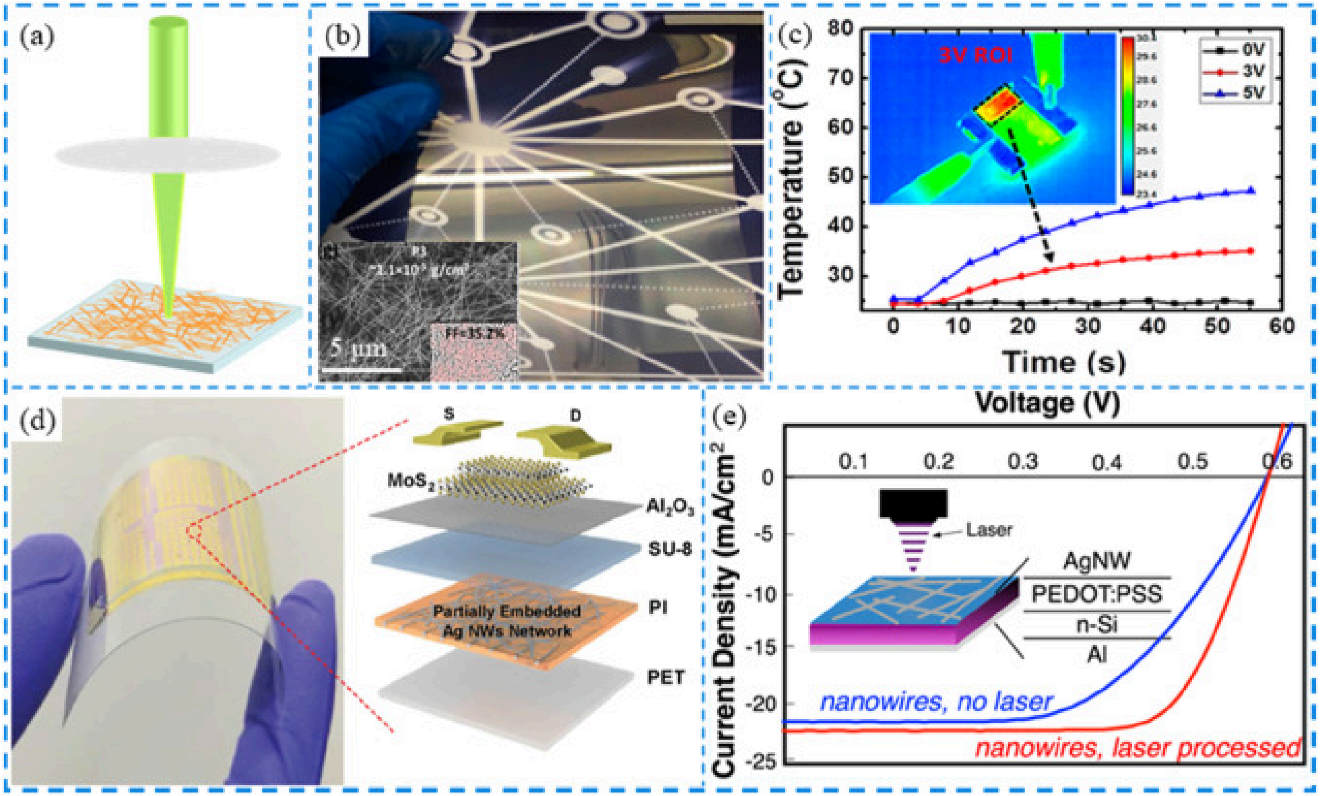

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Fig. 21. Applications of laser-induced joining of metallic nanowire networks as electrodes. (a) schematic showing laser-induced joining of metallic nanowire networks. (b) uniform printing of Ag nanowires on an 8-inch by 4-inch PET substrate by roll-to-roll printing followed by a laser-induced joining process. Inset, SEM image of printed Ag nanowire networks films on the PET substrate with an area density of ~1.1×10^{-5} g/cm^2. (c) laser processed Ag nanowire films on a PET substrate as a transparent heater. The temperature is measured as a function of time during continuous joule heating. (b, c) reproduced with permission from Ref. [129]. Copyright 2015, American Chemical Society. (d) photograph of a bent flexible MoS_2 transistor array. The right inset shows a representation of the layer-by-layer structure of the MoS_2 transistor. Reproduced with permission from Ref. [268]. Copyright 2016, Wiley-VCH. (e) comparison of *J-V* characteristic curve for a hybrid organic poly(3,4-ethylenedioxythiophene)

polystyrenesulfonate (PEDOT: PSS)/n-Si heterojunction cell before and after laser processing. Reproduced with permission from Ref. [219]. Copyright 2015, American Chemical Society.

Functional nanodevices

Laser joining can generate strong bonding between nanoscale materials and enhances the mechanical strength of structures used in a variety of functional devices. For example, a strong joint between graphene and a polymer substrate can be formed by laser scanning, which can then be used for simultaneous transferring and patterning of graphene structures simultaneously [248]. The strong bonding between graphene and a flexible polymer substrate formed by CW laser scanning is ideal for the fabrication of mechanically robust flexible supercapacitor devices. A device manufactured in this way has maintained its performance characteristics after ~10000 bending test [249]. More details on the laser joining of graphene and polymer substrates can be found in the earlier section entitled “Joining of two-dimensional materials” .

Laser-induced joining and sintering of metallic nanoparticles and nanowires, can also improve the adhesion between metallic nanoparticles [155,170,192], nanowires [25,223] and a glass or polymer substrate. This occurs because surface plasmon induced melting of the interface between the metallic nanoparticles or nanowires and the polymer substrate allows the nanoparticles/nanowires to become embedded in the substrate. This acts to increase the adhesion strength and improves the mechanical stability of a flexible device when it is subjected to a large number of bending cycles [76,79]. Alternatively, instead of direct laser sintering of metallic nanoparticles on the substrates, Zhou et al. [269] proposed that the Ag precursor solution can be directly written on the substrate allowing the synthesis and sintering of Ag nanoparticles to occur simultaneously. This process is facilitated by the addition of PVP to the precursor solution which promotes the enhanced adhesion of the Ag nanoparticle thin film to the substrate. The resulting patterned Ag nanoparticle thin film is resistant to the adhesive tape test over 50 applications showing that the film is strongly bonded to the substrate. Such devices also retain their sheet resistance after many bending cycles indicating that these laser-processed materials are excellent candidates for inclusion in practical wearable and flexible devices (Fig. 22(a, b)). In a similar process, the heat introduced by the excitation of surface plasmon resonances in metallic nanoparticles under laser irradiation can also be utilized for joining of bulk materials such as glass. In an example of this approach to bonding, a glass-metal composite with a 10 μm thick surface layer of embedded randomly distributed Ag nanoparticles is laser irradiated through a transparent glass plate. Heat generated by excitation of the surface plasmon resonance in the Ag particles is transferred from this layer into the overlying glass plate. This produces melting at the interface and the formation of a strong bond between the two surfaces (Fig. 22(c, d)) [270].

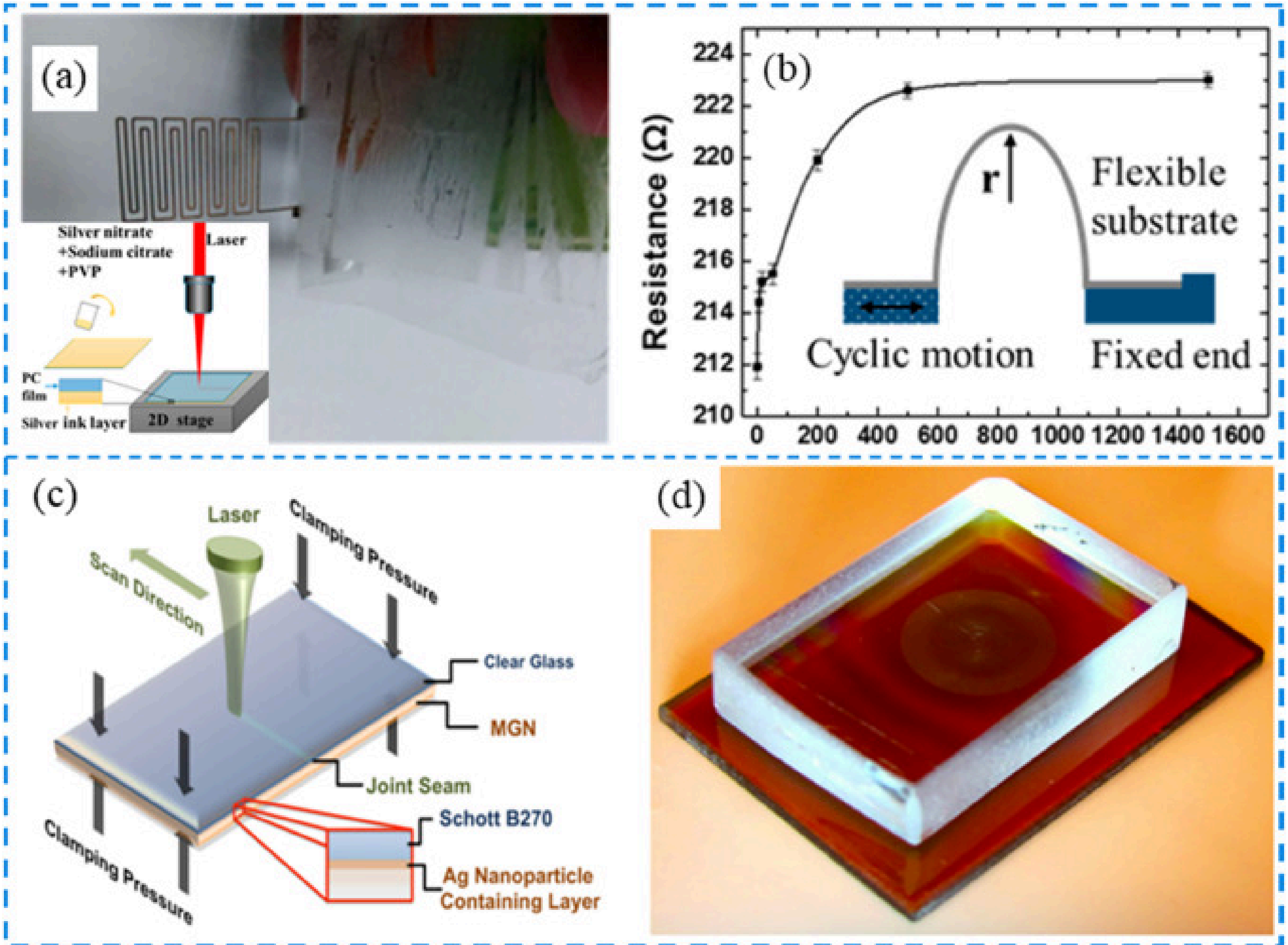

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Fig. 22. Laser joining induced strong mechanical strength in a number of devices. (a) Optical photograph of adhesion test for Ag nanoparticle thin film obtained by laser direct writing, left inset, schematic diagram for the fabrication process. (b) Resistance after bending for the Ag nanoparticle thin film for up to 1500 times. (a, b) Reproduced with permission from Ref. [269]. Copyright 2016, American Chemical Society. (c) Schematic diagram for laser-induced joining of glass plate to glass with layer of embedded Ag nanoparticles. (d) Photograph showing result of laser-induced joining of 4 mm clear glass-Schott B270 (on top) to glass

substrate (bottom) which has a layer of Ag nanoparticles embedded in a thin layer on the top surface. (c, d) Reproduced with permission from Ref. [270]. Copyright 2014, AIP Publishing.

Laser-induced nanojoining provides a straightforward way of obtaining robust mechanical contacts between different nanoscale building blocks. This technique is promising for the fabrication of functional circuits based on different nanoscale materials. Some systems studied to date include laser-induced joining of metal-semiconductor nanowire combinations such as Ag-TiO_2 [134], Ag-ZnO [135], Au-ZnO [241], Ag-CuO [138]. In each case, the electrical characteristics of the junction formed between the two materials are compared before and after laser-induced joining. For example, Xiao et al. demonstrated that the laser joining of a Ag-CuO nanowire heterojunction is accompanied by a reduction in the Schottky barrier at the Ag/CuO interface. The inter-diffusion of atoms at the interface due to prolonged laser irradiation time was found to decrease the contact resistance and promote charge transfer [138]. Focused laser irradiation has also been shown to enable the joining of semiconductor nanowires, such as CuO-CuO [233], ZnO-ZnO [151] and GaN-SnO_2 [242]. Therefore, it is of great interest to apply laser joining to the fabrication of nanowire based functional circuits with other electrical properties, such as p-n junctions and metal-oxide-semiconductor heterojunctions. By careful controlling the laser processing parameters it is expected that laser irradiation may have many applications in contact engineering of complex structures based entirely on nanoscale building blocks.

This laser-based technology will be especially applicable when nanoscale building blocks consisting of nanowires and 2D materials such as graphene are used as components in the manufacture of a range of different nanoscale devices. Laser-induced selective and localized heating can not only induce robust bonding between metal electrodes and nanowires/2D materials, but can also engineer specific interfacial nanostructures via the localization of plasmons. Research on laser-induced contact engineering of nanowires includes studies on In_2O_3[30,173], TiO_2 [136,256], ZnO [135], SiC [137,257], CNT [172], Cu [228] for implementation in memristor, transistor devices and sensing. Laser processing has also been applied to engineering of graphene-electrode contacts [133,250,251]. Some details for contact engineering of nanowires and graphene on electrodes by laser joining have been described in earlier sections in this review. Furthermore, by controlling the laser processing parameters, such as processing time, laser fluence and the composition of the ambient atmosphere, controlled engineering of the devices' performance can be realized. This suggests that laser-induced contact engineering of nanoscale building blocks on electrodes may be a promising technology for enabling new functionalities in nanoscale devices via a bottom-up approach.

Optical applications

Device containing Ag, Au and other metallic nanoparticles and nanowires are of much interest due to their intrinsic surface plasmon resonance properties and the way in which the excitation and propagation of plasmons can be used to couple to other physical phenomena. Two examples include the enhancement in fluorescent emission, and increasing Raman scattering and hyper-Raman scattering cross-sections for applications in molecular diagnostics, biomedical devices, and other nano-structured detectors [159]. Laser-induced nanojoining of nanoparticle and nanowire building blocks has been shown to provide a simple way of optimizing these optical properties [195].

Plasmonic resonance effects are dominant in surface-enhanced Raman spectroscopy (SERS) where metal nanostructures incorporating Ag or Au components can enhance the Raman signal by many orders of magnitude. SERS is then an ideal tool for the detection of trace chemicals in chemical, biomedical and environmental applications. The SERS enhancement factor does, however, depend on the size and morphology of the metallic nanoparticles so that engineering of plasmonic structures has proven to be particularly effective in optimizing the detectivity for a specific molecular component. While a spherical nanoparticle generally exhibits weak enhancement compared to an anisotropic particle, Nie et al. [271] demonstrated that anisotropic, high aspect ratio colloidal metal nanoparticles can further increase the SERS enhancement factor to as high as 10^{14} facilitating the detection of single molecules. Laser-induced joining of metallic nanoparticles into anisotropic nanostructures and their assembly into sub-micron assemblies provides a simple, non-contact method for enhancing the detection limit of SERS [159,167,187]. As an illustration of this effect, Hu et al. obtained a Raman signal from adenine at a concentration of 10^{-4}M using a substrate consisting of randomly oriented single Ag nanoparticles [166]. It was found that joining of Ag nanoparticles by fs laser irradiation improved this detection limit by a factor of 5-6 compared to the unprocessed substrate of randomly distributed single Ag nanoparticles (Fig. 23(a)). Simulation of the normalized electric field distribution for adjacent and joined 2 and 4 Ag nanoparticle structures with a gap of 5 nm and −5 nm, respectively, shows appreciable electric field enhancement in the ring-shaped hot spot in the necking region as well as in the outer periphery of the joined Ag nanoparticles (Fig. 23(b)) This enhancement can be associated with plasmon excitation and is responsible for the enhanced SERS cross-section in the laser processed substrate. This result demonstrates that a substrate with joined Ag nanoparticles is effective as a reliable SERS detector probe.

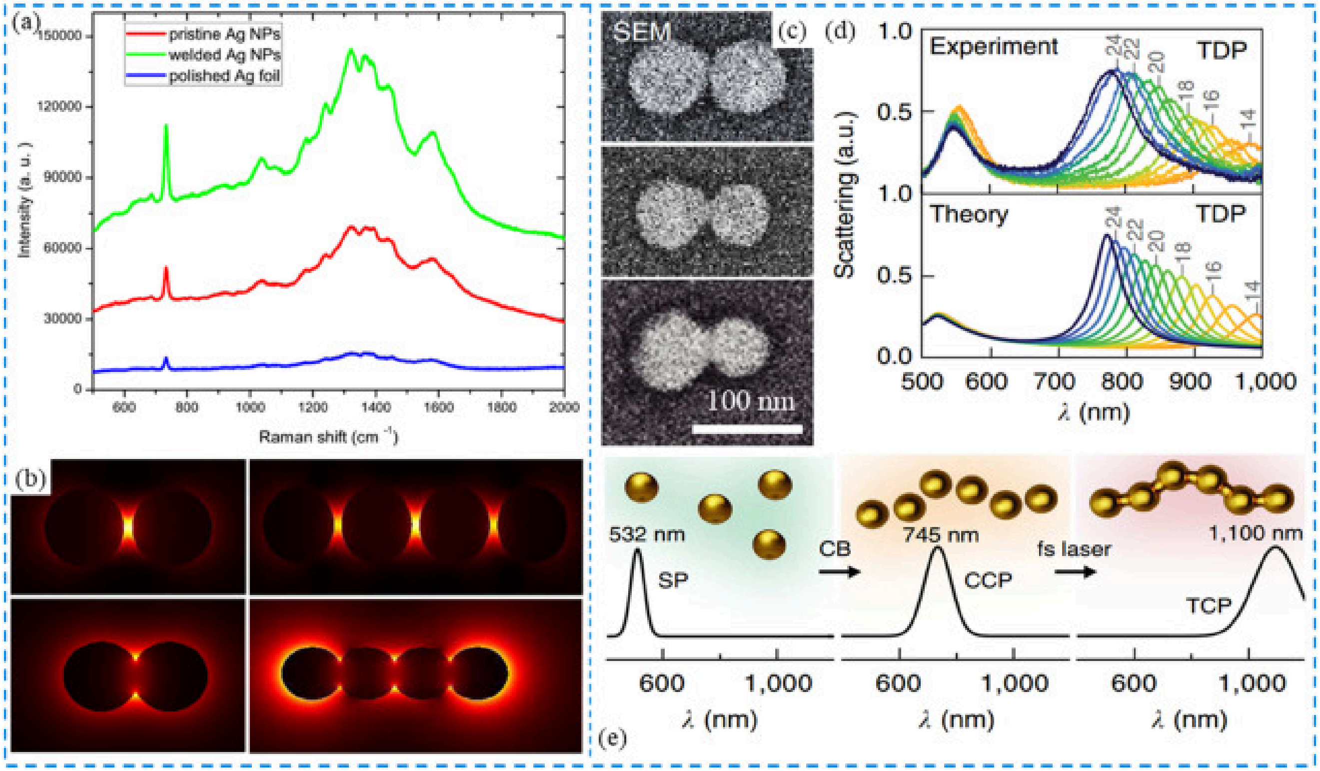

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Fig. 23. Laser-induced nanojoining for optical applications (a, b) Ag nanoparticle joining by laser for SERS application. (a) Raman signal showing the detection of adenine at a concentration of 10^{-4}M under for Ag foil, pristine Ag nanoparticles and laser joined Ag nanoparticle substrates. (b) Normalized electric field distribution for adjacent and joined Ag nanoparticles containing 2 and 4 particles, respectively. Reproduced with permission from Ref. [166]. Copyright 2012, AIP Publishing. (c-e) Au nanoparticles joined by laser demonstrating tunable optical properties. (c) SEM images of gold nanoparticle dimer with a gap of ~1 nm and bridges of ~15 nm and 30 nm (from top to bottom). (d) Optical scattering spectrum for the gold nanoparticle dimer for controlled thread width values (top: experimental results, bottom: simulation results). (e) Laser processing for strings of Au nanoparticles shows the appearance of new plasmon resonance peaks. (c-e) Reproduced with permission from Ref. [168]. Copyright 2014, The Nature Publishing Group.

Similarly, Qin et al. demonstrated that the CW laser irradiation induced joining and reshaping of Au nanoparticles could lead to the giant enhancement of photoluminescence signal [195]. This enhancement is attributed to the joining of adjacent Au nanoparticles in which the incident light field can be significantly enhanced at the joint area. Lin et al. further shows that the joining of branched Ag nanowires can lead to controlled engineering of light emission properties in perspective of incident polarization directions for the application of photonic logic gates in photonic devices [148]. It shows that the energy loss in the contact area for joined Ag nanowire is much smaller compared to the branched nanowires with a gap, therefore, leading to an increased switching ratio of the transmitted fields.

The optical response of joined nanoparticles is often quite different from that of the separated metallic nanoscale pairs due to changes in the distribution of surface charge. A difference in charge distribution can be associated with a change from the dipole-dipole interaction involving isolated nanoparticles to a conductivity coupled quadrupole interaction in particle agglomerates [272]. Hermann et al. demonstrated that ultrafast laser irradiation could be used to join of a large number of identical Au nanoparticles in a continuous string-like structure [168]. This generates new resonant plasmon absorption at a wavelength that depends on the number of nanoparticles in the chain. As the single plasmon resonance of individual gold nanoparticles with a diameter of 50 nm is located at 532 nm, linking these nanoparticles in solution with an organic bridge yields a new plasmon resonance peak at ~ 745 nm. This wavelength of this resonance moves to ~1100 nm when the Au nanoparticles are joined after fs irradiation to form a string-like structure as shown in Fig. 23(e). A similar result is obtained for the end-to-end joining of Au nanorods by fs laser irradiation [177], as discussed in the previous section about the joining of metallic nanoparticles. This study also found that the growth of the size of the necking region, or thread width, with controlled laser irradiation (Fig. 23(c)) induced a red-shift of the longitudinal mode in the optical scattering spectra, while the transverse mode was hardly affected (Fig. 23(d)). Other optical properties for the Au nanostructure could also be controlled via laser irradiation and were found to be dependent on nanoparticle size and the number of nanoparticles forming a single chain structure. This suggests that laser nanojoining of metallic nanoparticles can enable the assembly of a widely tunable plasmonic material with a well-defined optical response. In comparison with the direct fabrication of nanostructures with controlled gap distances by e-beam lithography for tailoring their plasmon resonance behavior [178,179,272], laser nanojoining provides a simple relatively low cost way for controlled tuning of optical properties. In addition, the laser-based method lends itself to the creation of complex structures (e.g. rings and chiral strings) based on fabricated arrays of joined nanoparticles.

While this laser irradiation technology has been widely applied to tailoring the optical properties of metallic nanoscale materials, the same approach can be used for engineering the optical response of semiconductor nanomaterials. For example, Xing et al.

demonstrated that ZnO-ZnO nanowires joined by focused fs laser irradiation demonstrated an improved photocurrent when excited with visible light [151]. This improvement arises when laser irradiation causes the generation of oxygen vacancies that function as defect levels in the ZnO bandgap enhancing optical absorption in the visible spectrum. It is expected that the joining of other nanoscale materials such as different combinations of semiconductor nanowires, metal-semiconductor nanoparticle/nanowire heterojunctions, and composites of metal nanoparticle-2D ultrathin materials will exhibit unique optical and optoelectrical properties.

Conclusion and outlook

In this review, we highlighted current progress in the field of nanojoining, especially laser nanojoining of a variety of different nanoscale materials. We discussed the various methods that have been used in laser nanojoining and the application of these techniques for processing of a wide range of materials ranging from zero-dimensional nanoparticles, one-dimensional nanowires, two-dimensional ultrathin materials to the hybrid combination of nanomaterials with different dimensions. Tailoring of laser processing in practical systems for potential applications is also outlined. Laser processing of these nanomaterials has been shown to lead to the formation of robust mechanical joints between nanoscale building blocks, permitting the assembly of these units into functional devices. In many cases, laser processing is also found to significantly decrease contact junction resistance and facilitate engineering of atomic structure allowing the optimization of interface properties and the creation of complex nanostructures. We discuss work that has been carried out on the engineering of laser processed materials for use in the fabrication of electrodes, functional nanodevices, and new optical applications. Some promising directions for future research in this field can now be identified. These are summarized as follows:

(1) Laser nanojoining of different types of nanomaterial and combinations of nanomaterials for the production of heterojunctions for use in the next generation of nanodevices. Examples would include the joining of semiconductor nanomaterials such as Si, Ge, GaAs, etc., wide band-gap insulating nanomaterials such as SiO_2, and Al_2O_3, joining of ceramic and polymer nanomaterials and exploration of the possibility of forming heterojunctions of these materials with other compositions by laser joining. Joining of nanomaterials in complex structures incorporating a variety of different types of material by laser processing is required for the creation of circuits with sophisticated functionalities as found in NEMS applications. More work is needed on the laser tailoring of interfaces between nanoscale materials with different dimensionalities such as nanoparticle-nanowire, nanoparticle-2D material and nanowire-2D material heterojunctions. This work

would be driven by developing applications for nanosystems in sensing, optoelectronics, photoelectronic, photocatalytic, solar cells, and bioelectronics. For example, partially or fully embedding the plasmonic metallic nanoparticles into semiconductor nanoparticles/nanowires can significantly improve the photocatalytic behavior compared to that just contact the metallic nanoparticles on the surface due to the increased interface area that hot electron from plasmonic metallic nanoparticles can transfer [216]. Laser nanojoining can be a perfect method for forming this type of heterojunction for these applications.

(2) Laser processing induced interface engineering for practical device applications. This includes research on the optimization of contact engineering between semiconductor nanowires on electrodes as well as laser-induced interface engineering of other nanomaterials such as semiconductor and insulator nanowires and 2D materials. A thorough understanding is needed on how laser processing affects the lattice structure and the bonding of atoms at the interface between different nanomaterials and how this determines carrier and thermal transport across this interface. This information is fundamental if nanoscale building blocks are to be assembled for use in practical devices. We need increased knowledge of laser irradiation effects in nanomaterials, with and without plasmonic interactions, to establish how these interactions can be used for tuning of device performance [[273], [274], [275], [276], [277], [278]].

(3) Research on laser mediated joining mechanisms. The interaction between laser radiation with nanoscale materials requires further understanding of dynamic processes such as lattice heating, surface atom migration and the driving force behind these phenomena. To date, the characterization of these effects has focused primarily on metallic nanomaterials. In-situ laser-coupled TEM characterization will provide an ideal tool for the dynamic study of the laser nanojoining process and should be applied to future research on laser joining of different types of nanoscale materials [214]. In-situ monitoring by optical extinction spectroscopy provides an alternative method for the characterization of the laser-induced joining process of large aggregates of metallic nanomaterials [168,177]. Theoretical studies, based on molecular dynamic simulations, will also lead to a better understanding of atomic configurations in laser processed nanomaterials and can be used to support experimental observations of the nanojoining process [129,[279], [280], [281], [282], [283], [284]]. The development of novel ways to measure the temperature distribution following laser-induced nanoscale joining of materials will provide a fuller picture of thermodynamic conditions during the laser-induced nanojoining process [128]. Numerical techniques that simulate the temperature distribution in

nanoscale building blocks under laser illumination in realistic laser joining protocols are necessary to complete our understanding of plasmonic effects involving metallic nanomaterials [126,175,181]. With the advancement in nanotechnology and the continuous development of affordable, more functional laser/laser systems, the laser nanojoining field will remain a hot topic for years to come.

CRediT authorship contribution statement

Ming Xiao: Conceptualization, Writing - original draft. **Shuo Zheng:** Writing - review & editing. **Daozhi Shen:** Writing - review & editing. **Walter W. Duley:** Supervision, Writing - review & editing. **Y. Norman Zhou:** Supervision, Writing - review & editing, Funding acquisition.

Declaration of Competing Interest

The authors report no declarations of interest.

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> …Among the existing reports of laser processing of integrated electrode, In et al. constructed monofilament fiber supercapacitors by a laser synthesized strategy [35]. However, in the water splitting electrolyzer area, the current reports mainly focused on the laser preparation of catalysts [36,37], and the construction of the integrated electrodes and electrolyzer by laser still needed further work. Herein, we presented the molybdenum carbide micropillars on Mo foil (Mo2C-L/Mo) by focused laser ablation in CH4 atmosphere at normal pressure and temperature, which realized the simultaneous formation of Mo2C and construction of micropillars.…

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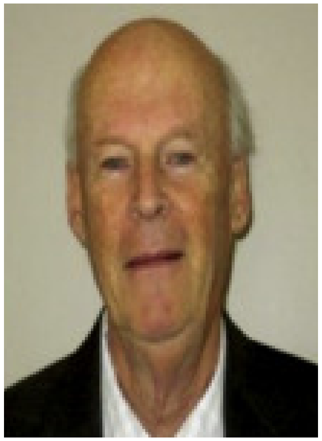

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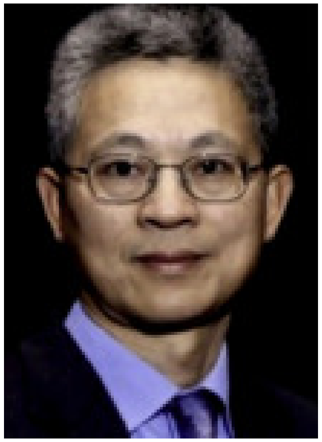

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